# Voice of Village: Fund Transparency & Issue Reporting System

Major project submitted in partial fulfillment of the requirement for the award of the degree of

**BACHELOR OF TECHNOLOGY**

**IN**

**COMPUTER SCIENCE AND ENGINEERING**

Under the esteemed guidance of

**CH. Sandhya**
**Assistant Professor**

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# CERTIFICATE

This is to certify that the B.Tech Major Project report entitled **"Voice of Village: Fund Transparency & Issue Reporting System"** is a bonafide work done by Name of the student **P.Uday - 22R11A05M9, V.Satwik Reddy - 22R11A05N9, M. Manikanth Reddy - 22R11A05M0** in partial fulfillment of the requirement of the award for the degree of Bachelor of Technology in "**Computer Science and Engineering**" from Jawaharlal Nehru Technological University, Hyderabad during the year 2025-2026.

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We, **P. Uday , V. Satwik Reddy, M. Manikanth Reddy**, bearing Roll Nos. **22R11A05M9, 22R11A05N9, 22R11A05M0**, hereby declare that the project seminar report entitled **"Voice of Village: Fund Transparency & Issue Reporting System"** is done under the guidance of **Ms.Ch.Sandhya**, **Assistant Professor**, Department of Computer Science and Engineering, Geethanjali College of Engineering and Technology, is submitted in partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology in Computer Science and Engineering**.

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# ABSTRACT

In recent times, improving transparency and accountability in rural governance has become a critical concern, particularly in the management of public funds and resolution of local issues. Conventional approaches often rely on manual processes and lack proper tracking systems, resulting in delays, inefficiency, and limited communication between citizens and authorities.

The proposed system, *Voice of Village: Fund Transparency & Issue Reporting System*, is a web-based application developed to address these challenges by enabling structured interaction between villagers and governing bodies. The platform allows users to report local issues such as infrastructure damage, sanitation problems, and service disruptions, while also providing authorities with tools to monitor, manage, and resolve these complaints efficiently.

The system incorporates role-based access control, supporting different users including villagers, administrators, Sarpanch, and government officials. Each role is assigned specific functionalities to ensure secure and organized usage. Users can submit complaints with relevant details, track their progress, and engage through feedback mechanisms, promoting better communication and transparency.
Additionally, the platform enhances visibility into the allocation and utilization of village development funds. By making such information accessible, the system helps reduce misuse of resources and strengthens trust among stakeholders.

The application is built using modern web technologies, with React.js for the frontend, Node.js and Express.js for backend operations, and MySQL for data management. Overall, the system provides a scalable and efficient solution that supports transparent governance, faster issue resolution, and increased citizen participation in rural development.

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# LIST OF ABBRIVATIONS

**PDS** – Public Distribution System

**NGO** – Non-Governmental Organization

**CSR** – Corporate Social Responsibility

**SDG** – Sustainable Development Goals

**GP** – Gram Panchayat

**FG** – Fund Governance

**IR** – Issue Reporting

**VMF** – Village Management Fund

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# 1. Introduction

## 1.1 Overview of the Project

With the rapid growth of digital technologies, there is a rising need to modernize governance systems, especially at the village level. Traditional methods used for managing public issues and monitoring fund usage are often manual, unorganized, and lack proper tracking mechanisms. This leads to delays, poor communication, and limited awareness among citizens regarding decision-making processes and resource allocation.

The *Voice of Village: Fund Transparency & Issue Reporting System* is developed as a centralized web-based platform to overcome these challenges. It connects villagers with local authorities and provides a structured way to report and manage issues. Citizens can submit complaints related to infrastructure, sanitation, water supply, electricity, and other public services through a digital interface.

Each reported issue is stored with detailed information such as title, description, category, and optional image proof. Authorized users like Sarpanch, administrators, and officials can review these complaints, update their status, and take necessary actions. The system also allows users to track progress and communicate through comments, improving clarity and responsiveness.
A key feature of the platform is its ability to provide transparency in fund utilization. It offers insights into how village development funds are allocated and spent, helping reduce misuse and increasing accountability. Interactive dashboards and visual representations further simplify complex data for easy understanding by users.

The application is designed using modern web technologies, including React.js for the frontend, Node.js and Express.js for backend processing, and MySQL for database management. Overall, the system acts as a digital bridge between citizens and authorities, enabling efficient issue resolution and transparent governance.

## 1.2 Problem Statement

Achieving transparency and accountability in village-level governance continues to be a significant challenge, particularly in developing regions where digital adoption is still evolving. Most existing systems used for handling public issues and monitoring fund utilization rely heavily on manual processes or partially digitized methods. These approaches lack proper structure, real-time tracking, and reliable documentation, which often results in delays, inefficiency, and poor service delivery.

Villagers frequently encounter difficulties when attempting to report local problems such as damaged roads, irregular water supply, sanitation issues, and electricity failures. In the absence of a dedicated platform, complaints are usually communicated through informal channels like verbal requests or handwritten applications. This not only makes the process inconsistent but also increases the chances of complaints being ignored, misplaced, or delayed.

Another critical issue is the absence of a centralized system that connects citizens directly with local authorities. Without a unified platform, communication remains fragmented, leading to misunderstandings and lack of coordination among stakeholders such as villagers, Sarpanch, administrators, and government officials. Furthermore, there is no effective mechanism to track the status of reported issues, leaving citizens uncertain about whether any action has been taken.

Transparency in the utilization of public funds is also a major concern. In many cases, citizens have little to no access to information regarding how funds are allocated, distributed, or spent on development activities. This lack of visibility creates opportunities for mismanagement and reduces public trust in governance systems.

In addition to these challenges, many existing solutions are not designed with user accessibility in mind. Complex interfaces, lack of localization, and limited awareness make such systems difficult to adopt in rural environments, especially for users with minimal technical knowledge.

Therefore, there is a clear need for an integrated, user-friendly, and transparent system that enables structured issue reporting, real-time tracking, effective communication, and clear visibility into fund utilization. Such a system should bridge the gap between citizens and authorities, improve administrative efficiency, and promote trust and accountability in village governance.

## 1.3 Objectives of the Project

The primary aim of the *Voice of Village: Fund Transparency & Issue Reporting System* is to design and implement a digital platform that enhances transparency, accountability, and efficiency in village-level governance. The system is intended to empower citizens by providing them with a structured and accessible way to report issues and stay informed about administrative activities.

The specific objectives of the project are as follows:

- To develop a web-based application that enables villagers to report local problems such as infrastructure damage, water supply issues, sanitation concerns, and other public service-related complaints in a systematic manner.
- To provide a transparent mechanism for tracking the status of reported issues, allowing users to monitor progress from submission to resolution.
- To implement a role-based access control system that supports multiple stakeholders, including villagers, Sarpanch, administrators, and government officials, ensuring that each user interacts with the system according to their responsibilities.
- To improve communication between citizens and authorities through features such as comments, updates, and feedback on reported issues.
- To promote transparency in the utilization of public funds by providing clear and accessible information about fund allocation and expenditure.
- To reduce delays, inefficiencies, and dependency on manual processes by digitizing the issue reporting and management workflow.
- To maintain a centralized and well-structured database for storing and managing all information related to users, issues, and fund activities.
- To ensure system security through proper authentication and authorization mechanisms, protecting user data and preventing unauthorized access.
- To design a scalable and modular system architecture that allows future enhancements such as notifications, analytics, and mobile application support.
- To strengthen trust and accountability between citizens and governing authorities by creating a transparent and interactive governance environment.

Overall, the project aims to bridge the communication gap between rural communities and administrative bodies by leveraging technology to create a more responsive, efficient, and participatory governance system.

## 1.4 Scope of the Project

The scope of the *Voice of Village: Fund Transparency & Issue Reporting System* focuses on developing a web-based platform that enhances communication,

transparency, and efficiency in village-level governance. The system is designed to provide a unified environment where citizens and authorities can interact in a structured and organized manner.

The platform covers the complete lifecycle of issue management, beginning from the submission of complaints by villagers to their resolution by responsible authorities. Users can report issues by providing relevant details such as title, description, category, and optional image evidence. These issues are then processed by authorized users, including the Sarpanch, administrators, and government officials, who can review, update status, and take appropriate actions.

The system also allows users to monitor the progress of their complaints in real time and participate in discussions through comments and feedback. This continuous interaction improves communication and ensures that users remain informed about the status of their issues.

In addition to issue management, the project includes a module for enhancing transparency in fund utilization. The platform provides visibility into how village development funds are allocated and used, enabling citizens to stay informed and encouraging accountability among authorities.

The system supports multiple user roles, each with specific permissions defined through a role-based access control mechanism. This ensures secure access and organized functionality for different stakeholders. Authentication and authorization features are implemented to protect user data and maintain system integrity.

From a technical perspective, the project includes the development of a backend using Node.js and Express.js for handling business logic and API communication, along with a MySQL database for efficient data storage. The frontend is built using React.js to deliver a responsive and user-friendly interface.

However, the scope of the project is limited to a controlled environment. It does not include integration with external government systems, large-scale deployment, or advanced features such as real-time notifications, geolocation-based tracking, or dedicated mobile applications. These aspects can be considered as potential enhancements in future work.

Overall, the project demonstrates a scalable and practical approach to improving governance at the village level by combining issue reporting and fund transparency within a single digital platform.

## 1.5 Methodology / SDLC Model Adopted

The development of the *Voice of Village: Fund Transparency & Issue Reporting*

*System* follows the Agile Software Development methodology. This approach is selected due to its flexibility, iterative nature, and ability to adapt to changing requirements during the development process. Since the project consists of multiple interconnected modules such as authentication, issue reporting, fund transparency, and role-based access, Agile provides an efficient way to develop and refine each component step by step.

Agile methodology divides the development process into smaller cycles known as sprints. Each sprint includes activities such as planning, design, development, testing, and review of specific features. This iterative approach allows the team to identify issues early, incorporate feedback, and continuously improve the system throughout its development.

One of the key advantages of using Agile in this project is its support for continuous feedback and incremental improvement. As each module is developed, it is tested and evaluated, ensuring better quality and reducing the chances of major errors in later stages. This approach also allows parallel development of frontend and backend components, improving efficiency and reducing development time.

The methodology emphasizes flexibility, enabling the addition or modification of features such as fund tracking, reporting mechanisms, and user interaction modules without disrupting the overall system. Regular testing at each stage ensures system reliability, stability, and correctness.

The overall development process follows these steps:

1. Requirement analysis and understanding of system needs
2. Planning and division of tasks into manageable modules
3. Iterative development of features such as user authentication, issue reporting, and database integration
4. Continuous testing and debugging of individual components
5. Integration of frontend and backend systems
6. Review and refinement based on feedback
7. Deployment and further improvements
8. By adopting the Agile methodology, the project ensures a structured yet adaptable development process, resulting in a reliable, scalable, and user-friendly system that meets the intended objectives effectively.

## 1.6 Organization of the Report

This report is organized into multiple chapters, each focusing on a specific aspect

of the *Voice of Village: Fund Transparency & Issue Reporting System*, providing a clear and systematic understanding of the project from concept to implementation.

**Chapter 1: Introduction**: This chapter presents an overview of the project, including the problem statement, objectives, scope, and the methodology adopted for development.

**Chapter 2: Literature Survey**: This chapter reviews existing systems related to issue reporting and governance transparency. It discusses their limitations, the need for the proposed system, and provides a comparative analysis.

**Chapter 3: System Analysis**: This chapter focuses on the feasibility study, including technical, economic, and operational feasibility. It also includes the Software Requirements Specification (SRS) and identifies functional and non-functional requirements of the system.

**Chapter 4: System Design**: This chapter describes the overall system design, including system architecture, database design, UML diagrams, user interface design, and design standards followed.

**Chapter 5: Implementation**: This chapter explains the technologies used and provides details about module-wise implementation, code integration strategy, and coding standards followed during development.

**Chapter 6: Testing**: This chapter discusses the testing strategies adopted, including unit testing, integration testing, system testing, and quality assurance practices. It also includes test cases and bug tracking details.

**Chapter 7: Results and Discussion**: This chapter presents the output of the system with screenshots, analysis of results, and evaluation of system performance.

**Chapter 8: Conclusion and Future Scope**: This chapter summarizes the work done, highlights the limitations and challenges faced, and suggests possible future enhancements.

**Chapter 9: References**: This chapter lists all the sources referred to during the project, including technical papers, websites, and other materials.

**Chapter 10: Appendices**: This chapter includes supporting documents such as SRS forms, test reports, project timeline, ethical considerations, plagiarism report, source code links, and other relevant materials.

This structured organization ensures that the report is comprehensive, logically arranged, and easy to understand for readers and evaluators.

# 2. Literature Survey

## 2.1 Review of Existing System

Existing systems in the domain of governance and public issue management primarily focus on either complaint registration or administrative record-keeping, but rarely provide a complete and integrated solution. Traditional approaches in villages largely rely on manual methods such as written applications, in-person reporting to local authorities, or informal communication channels. These methods are time-consuming, lack proper documentation, and often result in delays or loss of information.

In recent years, some digital platforms and e-governance systems have been introduced to improve communication between citizens and authorities. These systems allow users to submit complaints online and enable officials to manage and track them. Such platforms help in organizing data, improving response time, and maintaining records of reported issues. However, their implementation at the village level is still limited, and many of these systems are designed for urban or large-scale administrative use.

Issue tracking systems used in organizations also provide structured workflows for reporting, assigning, and resolving problems. These systems include features such as status updates, comments, and notifications, which enhance coordination among users. While they are efficient in controlled environments, they are not specifically designed to address the unique challenges of rural governance, such as accessibility, simplicity, and multi-role participation.

Additionally, some transparency initiatives focus on publishing government data related to budgets and expenditures. These systems aim to improve accountability by making financial information available to the public. However, they often lack user-friendly interfaces and do not provide real-time or detailed insights into how

funds are utilized at the village level.

Overall, existing systems offer partial solutions by addressing specific aspects such as issue reporting or data transparency. However, they operate independently and lack integration, making it difficult to achieve complete transparency, efficient issue resolution, and effective communication among all stakeholders in a village ecosystem.

## 2.2 Limitations of Existing Approaches

Despite the availability of various systems for issue reporting and governance management, several limitations still exist in current approaches, especially at the village level.

One of the primary limitations is the lack of integration. Most existing systems focus on a single aspect, such as complaint registration or data publication, without providing a unified platform that combines issue tracking, communication, and fund transparency. This results in fragmented data and inefficient coordination among stakeholders.

Another major drawback is the dependence on manual or semi-digital processes. In many villages, issues are still reported through written applications or verbal communication, which are not properly recorded or tracked. This leads to delays, misplacement of complaints, and lack of accountability in resolving issues.

Existing digital systems often lack transparency and verification mechanisms. Information provided by authorities is not always accessible or easy to understand for citizens, and there is limited visibility into how decisions are made or how funds are utilized. This creates a gap between citizens and governing bodies, reducing trust.

Additionally, most systems do not provide real-time tracking of issues. Once a complaint is submitted, citizens are often unaware of its progress or status. The

absence of status updates and feedback mechanisms limits user engagement and satisfaction.

Another limitation is the lack of role-based interaction. Many systems do not clearly define responsibilities for different stakeholders such as villagers, Sarpanch, administrators, and government officials. This results in confusion, overlapping tasks, and inefficient handling of issues.

Furthermore, user accessibility and interface design are often neglected. Existing platforms may be complex, not mobile-friendly, or not tailored for users with limited technical knowledge, making them difficult to use in rural environments.

Finally, transparency in fund utilization is either limited or completely absent in most systems. Citizens do not have access to clear and detailed information about how public funds are allocated and spent, which increases the risk of misuse and corruption.

These limitations highlight the need for a more comprehensive, user-friendly, and integrated system that can effectively address issue reporting, tracking, communication, and fund transparency in a single platform.

## 2.3 Need for the Proposed System

The limitations of existing systems highlight the urgent need for a comprehensive and integrated solution that can improve transparency, accountability, and efficiency in village-level governance. A system is required that not only allows citizens to report issues but also ensures proper tracking, communication, and resolution in a structured and reliable manner.

There is a strong need for a centralized platform where all stakeholders—villagers, Sarpanch, administrators, and government officials—can interact within a single system. Such a platform should eliminate fragmented communication and provide a unified view of all reported issues and administrative actions.

Another important requirement is the ability to track issues in real time. Citizens should be able to monitor the status of their complaints from submission to resolution. This increases transparency, keeps users informed, and encourages authorities to respond more efficiently.

The system must also incorporate a clear and structured mechanism for communication. Features such as comments, updates, and feedback are essential to ensure proper interaction between citizens and authorities. This helps in clarifying issues, providing updates, and improving overall coordination.

A key need is to enhance transparency in fund utilization. Citizens should have access to information regarding how public funds are allocated and used for development activities. This reduces the chances of corruption and builds trust in the governance system.

Additionally, there is a requirement for role-based access control to ensure that each stakeholder has clearly defined responsibilities and permissions. This helps in maintaining order, improving accountability, and preventing unauthorized actions.

The system should also be user-friendly and accessible, especially for users with limited technical knowledge. A simple and intuitive interface is necessary to encourage adoption and ensure effective usage in rural environments.

Therefore, the proposed *Voice of Village: Fund Transparency & Issue Reporting System* is needed to address these challenges by providing an integrated platform that combines issue reporting, real-time tracking, transparent fund management, and efficient communication, ultimately improving governance and citizen participation at the village level.

## 2.4 Comparative Study

A comparative analysis between existing systems and the proposed Voice of Village: Fund Transparency & Issue Reporting System highlights the

improvements and advantages offered by the proposed approach.

| **Feature** | **Existing Systems** | **EcoTrace System** |
|---|---|---|
| Issue Reporting | Available but often manual or unstructured | Structured digital issue reporting with detailed inputs |
| Issue Tracking | Limited or no tracking | Real-time status tracking (pending, in progress, resolved) |
| Data Integration | Fragmented across different systems | Centralized platform with unified data |

| | | |
|---|---|---|
| Communication | Minimal or informal communication | Comment-based interaction between users and authorities |
| Transparency | Limited visibility of processes | High transparency in issue handling and updates |
| Fund Utilization | Rarely included or not accessible | Clear visibility of fund allocation and usage |
| User Roles | Not clearly defined | Role-based access (villager, Sarpanch , admin, government) |
| Accessibility | Complex or not user-friendly | Simple, intuitive, and user-friendly interface |
| Accountability | Low due to lack of tracking | Improved accountability through tracking and records |

**Table 2.4 Comparative Study**

## 2.5 Summary

This chapter presented a detailed study of existing systems related to issue reporting, governance management, and transparency mechanisms. It highlighted how traditional and current digital approaches provide partial solutions, focusing either on complaint registration, administrative processes, or data publication, but failing to deliver a complete and integrated system.

The limitations identified include lack of data integration, dependence on manual processes, absence of real-time tracking, poor communication mechanisms, limited

user accessibility, and lack of transparency in fund utilization. These issues reduce efficiency, accountability, and trust between citizens and authorities.

Based on these observations, the need for a unified and user-friendly system was established. The proposed *Voice of Village: Fund Transparency & Issue Reporting System* addresses these gaps by integrating issue reporting, tracking, communication, and fund transparency into a single platform. It provides role-based access, real-time updates, and improved interaction among stakeholders.

Overall, this chapter justifies the importance of the proposed system and sets the foundation for the subsequent chapters, which focus on system analysis, design, implementation, and evaluation.

# 3. System Analysis

## 3.1 Feasibility Study

The feasibility study evaluates the practicality and viability of developing and implementing the *Voice of Village: Fund Transparency & Issue Reporting System*. It examines whether the proposed system is technically possible, economically viable, operationally effective, and achievable within the given time and cost constraints.

**Technical Feasibility**

The proposed system is technically feasible as it is built using widely used and well-supported technologies. The backend is developed using Node.js and Express.js, which provide efficient handling of server-side operations and API development. The frontend is implemented using React.js, enabling a responsive and interactive user interface.

The system uses MySQL as the database for structured data storage and management. Authentication and authorization are implemented using JSON Web Tokens (JWT), ensuring secure access to the system. Additional features such as image uploads, issue tracking, and role-based access control are implemented using standard libraries and frameworks.

All required tools and technologies are readily available, open-source, and compatible with standard development environments. The system does not require specialized hardware and can run on general-purpose computers, making it technically practical to develop and deploy.

**Economic Feasibility**

The project is economically feasible as it primarily uses open-source technologies, which significantly reduce development and operational costs. Tools such as Node.js, React.js, and MySQL are freely available and widely supported.

The system does not require expensive infrastructure or proprietary software. It can be deployed on low-cost cloud platforms or local servers, minimizing hosting expenses.

Maintenance costs are also manageable due to the modular design and use of standard technologies.

Overall, the project can be developed within a reasonable budget, making it cost-effective for implementation at the village level.

**Operational Feasibility**

The system is operationally feasible as it is designed to be simple, user-friendly, and accessible to all stakeholders, including villagers with limited technical knowledge. The interface is intuitive, allowing users to easily report issues, track progress, and interact with authorities.

Role-based access ensures that each user—villager, Sarpanch, administrator, and government official—can perform only the actions relevant to their responsibilities. This improves system efficiency and reduces confusion.

The system enhances communication, reduces manual effort, and improves transparency, making it practical and beneficial for real-world usage.

**Time & Cost Estimation**

The development of the system is achievable within the given academic timeline by following an Agile methodology. The project is divided into modules such as authentication, issue reporting, database integration, and frontend development, which are implemented in phases.

Since the project uses open-source tools, the overall cost is minimal and mainly involves development time and basic infrastructure requirements. Proper planning and modular development ensure timely completion of the project.

Overall, the feasibility study confirms that the proposed system is practical,

cost-effective, and suitable for implementation, with the potential to significantly improve transparency and efficiency in village governance.

## 3.2 Software Requirements Specification (SRS)

The Software Requirements Specification (SRS) defines the overall requirements of the Voice of Village: Fund Transparency & Issue Reporting System. It describes the system's functionality, user interactions, and constraints, serving as a foundation for system design and implementation.

**1. System Overview**

The system is a web-based application designed to improve transparency and communication in village governance. It enables users to report issues, track their progress, and view fund utilization details. The platform supports multiple user roles, including villagers, Sarpanch, administrators, and government officials, each with specific functionalities.

**2. User Classes and Characteristics**

- **Villager/Citizen:** Can register, log in, report issues, view issue status, and add comments. Basic technical knowledge is sufficient.
- **Sarpanch (Village Head)**: Responsible for reviewing reported issues, updating their status, and managing local governance activities.
- **Administrator:** Manages users, monitors system activity, and ensures proper functioning of the platform.
- **Government Official:** Oversees issue resolution, monitors progress, and ensures accountability in governance processes.

**3. Functional Requirements**

- User registration and login with secure authentication
- Role-based access control for different users
- Issue reporting with details such as title, description, category, and optional image

- Viewing and tracking the status of reported issues
- Updating issue status by authorized users (pending, in progress, resolved)
- Adding and viewing comments on issues for communication
- Centralized storage and retrieval of issue data
- Fund transparency module to display allocation and usage details
- Secure API communication between frontend and backend

**4. Non-Functional Requirements**

- **Performance:** The system should respond quickly to user requests and handle multiple users efficiently
- **Usability:** The interface should be simple, intuitive, and easy to use for all types of users
- **Security:** Data should be protected using authentication (JWT) and proper access control mechanisms
- **Scalability:** The system should support future enhancements and increased user load
- **Reliability:** The system should function consistently with minimal errors or downtime
- **Maintainability:** The code should be modular and easy to update or extend

**5. System Constraints**

- The system is dependent on internet connectivity for operation
- It is designed for a controlled environment and does not integrate with external government databases
- Advanced features such as real-time notifications and mobile applications

are not included in the current scope

**6. Assumptions and Dependencies**

- Users have access to a device with internet connectivity
- Basic knowledge of using web applications is assumed
- The system relies on standard technologies such as Node.js, React.js, and MySQL

Overall, the SRS provides a clear understanding of system requirements and serves as a blueprint for designing and developing the application effectively.

## 3.3 Functional and Non-Functional Requirements

This section describes the detailed functional and non-functional requirements of the Voice of Village: Fund Transparency & Issue Reporting System. These requirements define what the system should do and how it should perform.

**Functional Requirements**

Functional requirements specify the core functionalities that the system must provide.

- **User Registration and Login:**

  The system should allow users to register and login securely using email and pass credentials**.**

- **Role-Based Access Control:**

  The system should support multiple roles such as villager, Sarpanch, administrator, and government official, with specific permissions for each role.

- **Issue Reporting:**

  Users should be able to submit issues with details such as title,

description, category, and optional image upload.

- **Issue Tracking:**

  The system should allow users to view and track the status of their reported issues (e.g., pending, in progress, resolved).

- **Issue Management:**

  Authorized users should be able to update the status of issues and manage their resolution process.

- **Comment System**:
  Users should be able to add comments to issues to facilitate communication and provide updates.

- **Data Management**:

  The system should store and retrieve all data related to users, issues, and comments in a structured database.

- **Fund Transparency (Conceptual Module):**

  The system should provide information about fund allocation and usage to improve transparency.

- **API Integration**:
  The frontend should communicate with the backend through RESTful APIs.

**Non-Functional Requirements**

Non-functional requirements define the quality attributes and constraints of the system.

- **Performance:**
  The system should provide fast response times and handle multiple user requests efficiently.

- **Usability:**

  The interface should be simple, intuitive, and easy to navigate, even for users with minimal technical knowledge.

- **Security:**

  The system should ensure secure authentication using JWT and protect sensitive user data from unauthorized access.

- **Scalability:s**

  The system should be designed to handle future growth in users and additional features.

- **Reliability:**

  The system should operate consistently without frequent failures or errors.

- **Maintainability:**

  The codebase should be modular and well-structured to allow easy updates and maintenance.

- **Compatibility:**

  The system should work across different devices and modern web browsers.

Overall, these requirements ensure that the system is not only functionally complete but also efficient, secure, and user-friendly, meeting the needs of both citizens and authorities.

# 4. System Design

## 4.1 System Architecture

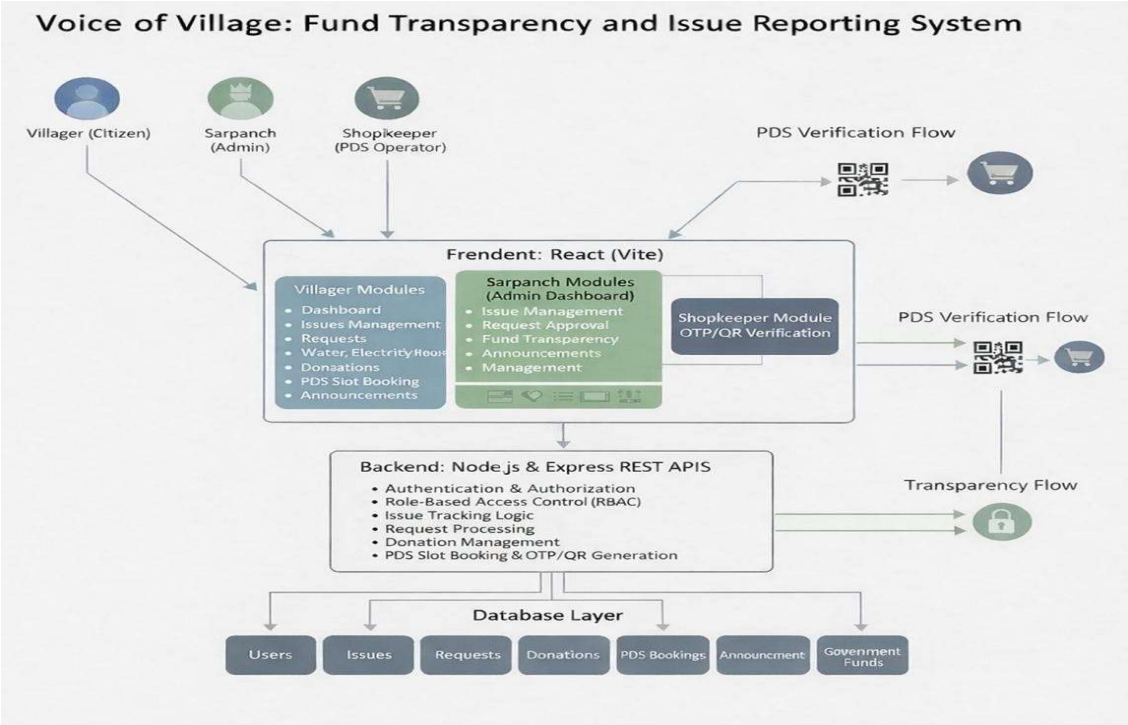


**Fig 4.1.1 System Architecture**

The *Voice of Village: Fund Transparency & Issue Reporting System* follows a three-tier architecture consisting of the frontend, backend, and database. This structure ensures proper separation of concerns and efficient system operation.

The frontend is developed using React.js and provides a user-friendly interface for interaction. The backend is built using Node.js and Express.js, which handle business logic, API requests, and authentication. The database uses MySQL to store and manage all system data such as users, issues, and comments.

The system follows a client-server model where the frontend communicates with the backend through APIs, and the backend interacts with the database to process and retrieve data. This architecture ensures scalability, maintainability, and smooth communication between system components.

## 4.2 Database Design (ER Diagram) / Data Flow Diagrams / Workflow

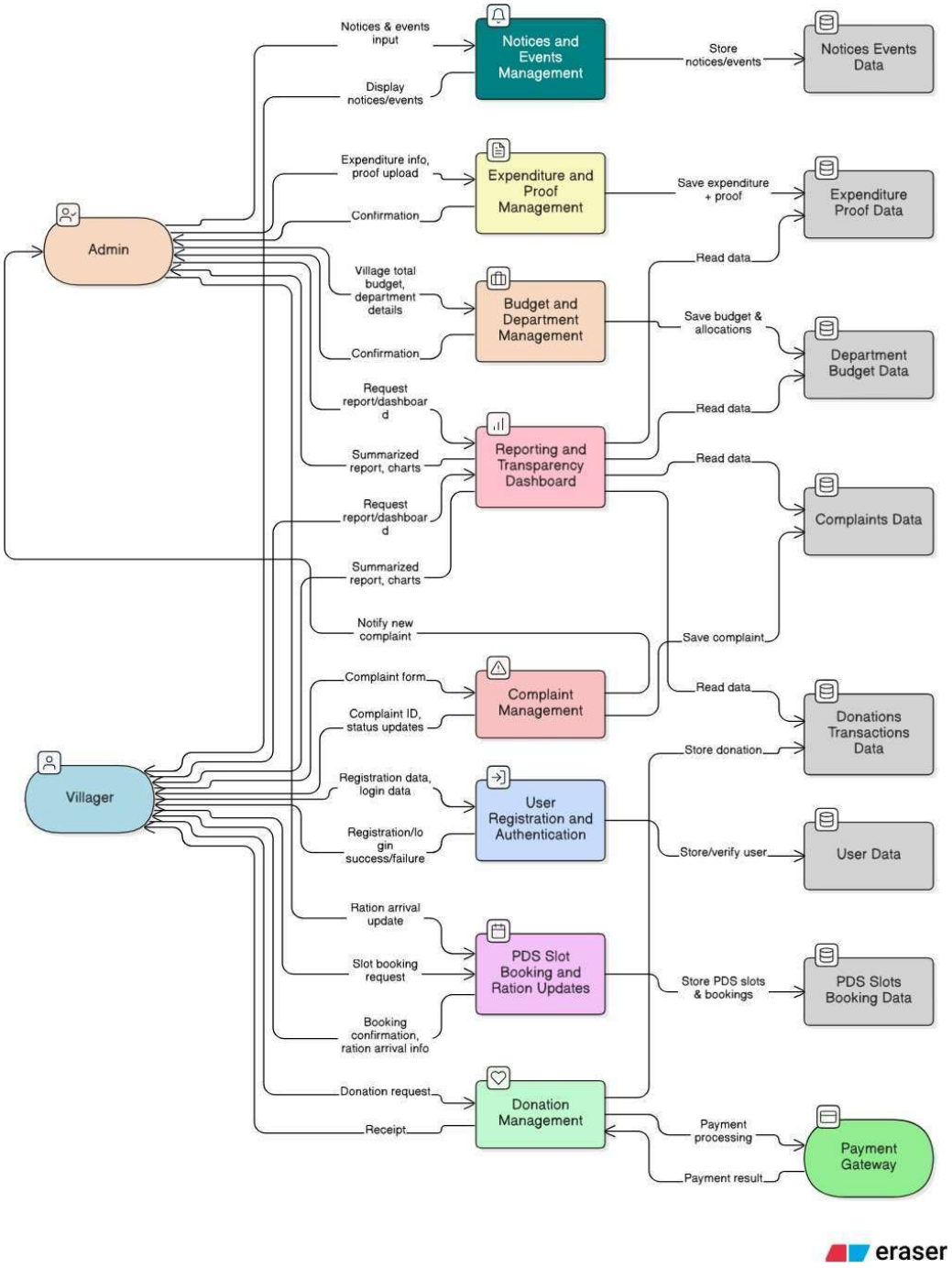


**Fig 4.2.1 ER Diagram**

The database design of the *Voice of Village: Fund Transparency & Issue Reporting System* is based on a relational model using MySQL. It is structured to efficiently store and manage data related to users, issues, and comments while maintaining data integrity and consistency.

The system consists of three main tables: **Users**, **Issues**, and **Issue Comments**.

The **Users** table stores user-related information such as name, email, password, role, and account creation details. It supports multiple roles including villager, Sarpanch, administrator, and government official.

The **Issues** table contains details of all reported issues, including title, description, category, status, image reference, and the user who submitted the issue. Each issue is linked to a user through a foreign key relationship.

The **Issue Comments** table stores comments made on issues by different users. It includes fields such as comment text, associated issue ID, user ID, and timestamp. This enables communication and discussion on each reported issue.

Relationships are established between these tables using primary and foreign keys to ensure referential integrity. The design avoids redundancy and supports efficient data retrieval and updates.

Overall, the database is designed to be simple, structured, and scalable, allowing easy management of data and supporting future enhancements.

## 4.3 UML Diagrams (Use Case, Class, Sequence, Activity)

### 4.3.1 Use Case Diagram

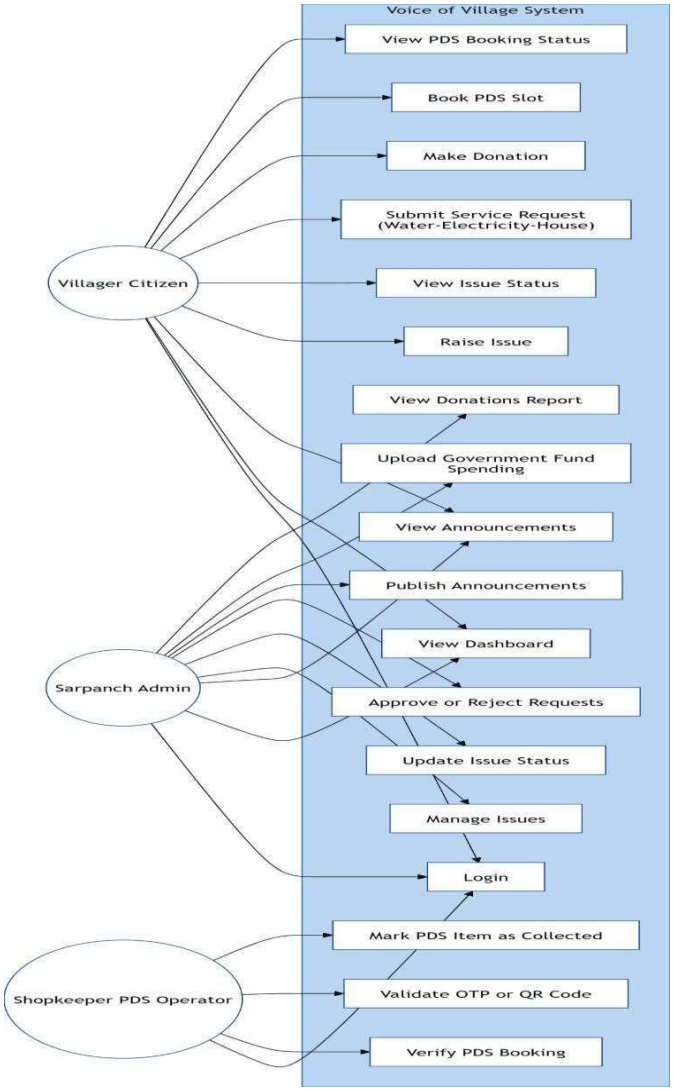


**Fig 4.3.1 Use Case Diagram**

The Use Case Diagram shows how different users interact with the system. The main actors are Villager, Sarpanch, Administrator, and Government Official.

Villagers can register, log in, report issues, view status, and add comments. Authorities like Sarpanch and officials can manage and update issues. Administrators oversee the system. It provides a simple view of system functionalities and user interactions.

### 4.3.2 Class Diagram

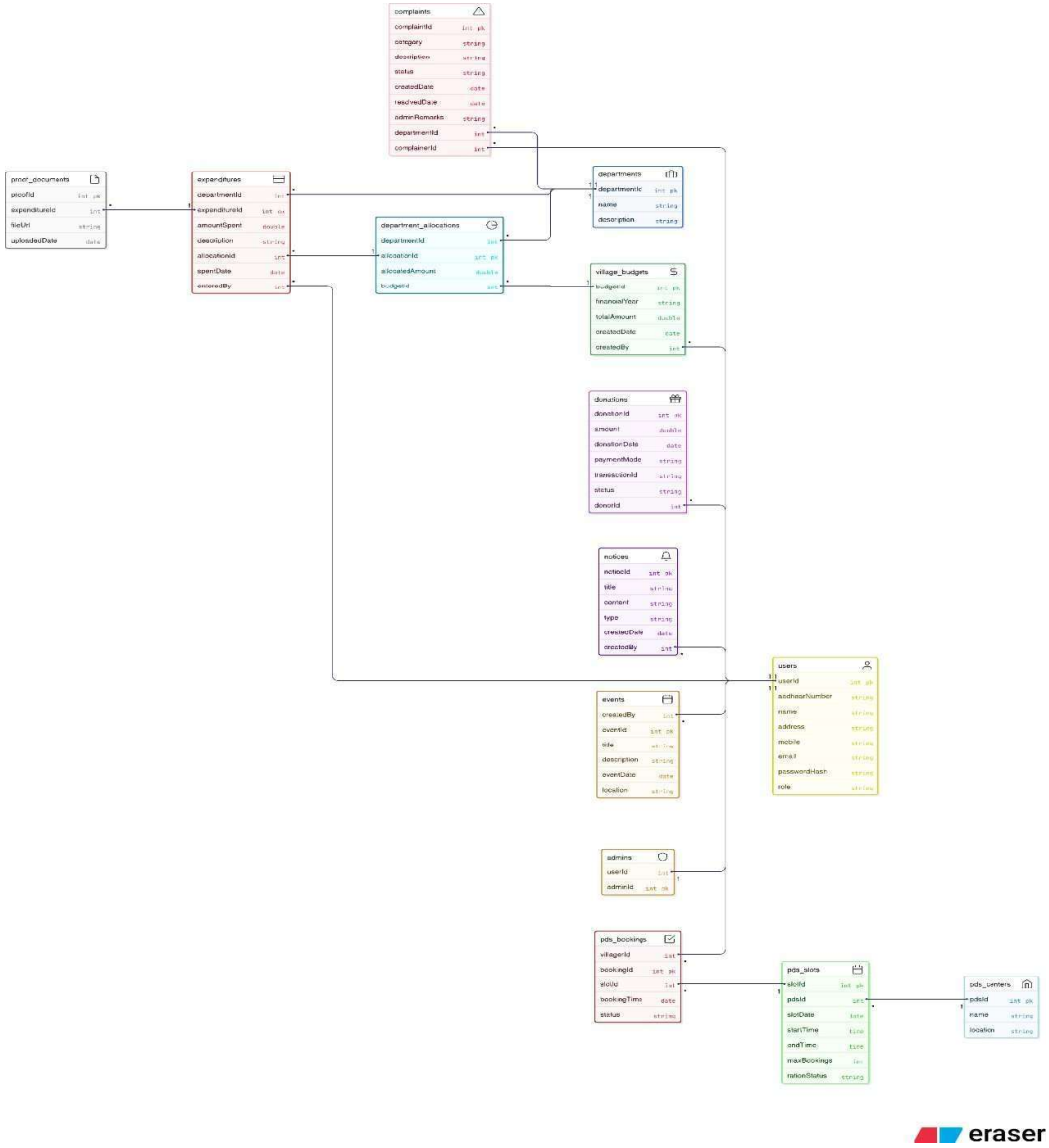


**Fig 4.3.2 Class Diagram**

The Class Diagram represents the structure of the system by showing different classes and their relationships. The main classes include User, Issue, and Comment. The User

class stores user details and roles. The Issue class contains information about reported issues. The Comment class manages user comments on issues. These classes are connected through relationships such as one user creating multiple issues and comments.

### 4.3.3 Sequence Diagram

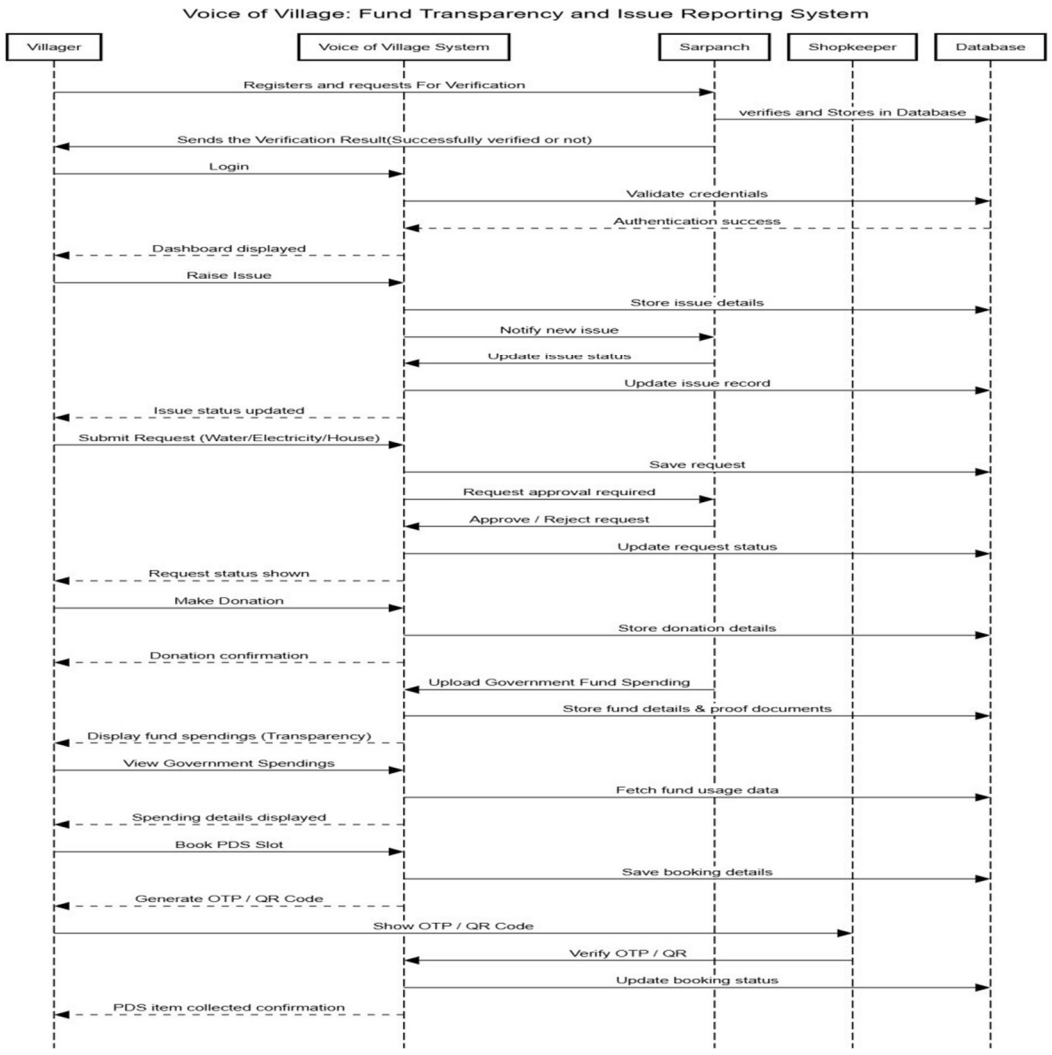


**Fig 4.3.3 Sequence Diagram**

The Sequence Diagram shows the flow of interactions between the user and the system over time. It illustrates how a user sends a request, the frontend forwards it to the backend, the backend processes it with the database, and then returns the response to the user. It helps in understanding the step-by-step execution of system operations.

### 4.3.4 Activity Diagram

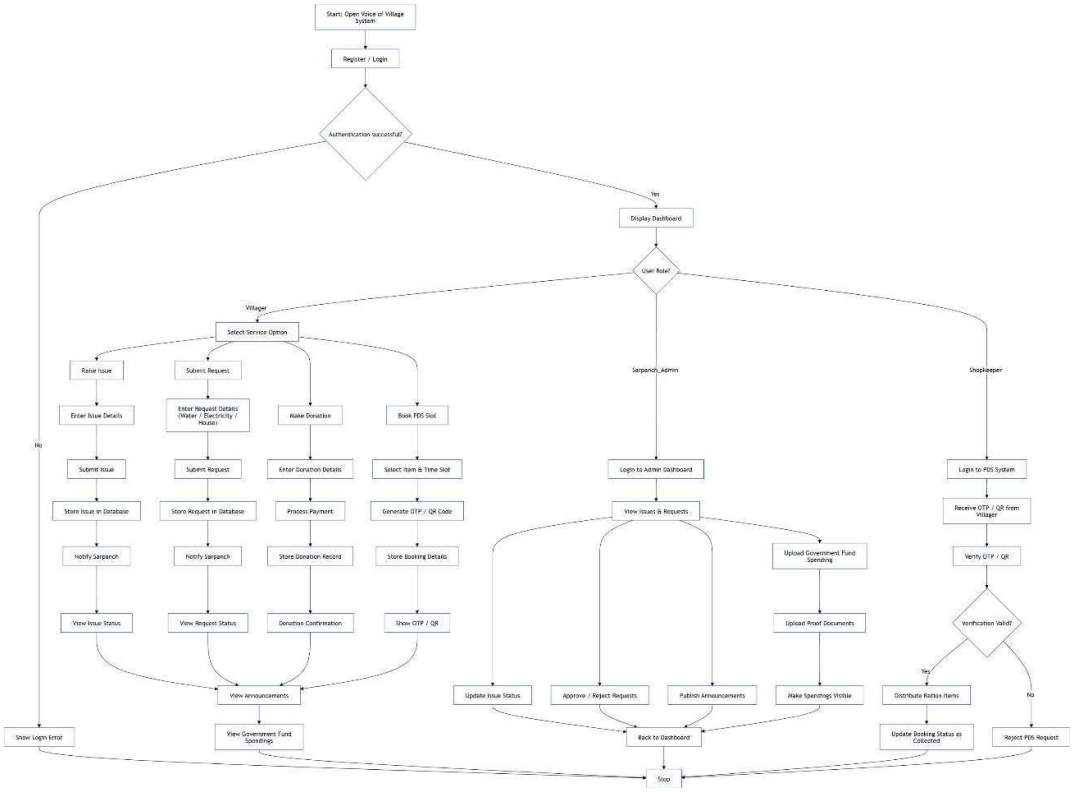


**Fig 4.3.4 Activity Diagram**

The Activity Diagram represents the workflow of the system. It shows the sequence of activities such as user login, issue submission, processing by authorities, and updating the issue status.

## 4.4 User Interface Design

**Fig 4.4 User Interface Design**

## 4.5 Design Standards Followed

The design of the *Voice of Village: Fund Transparency & Issue Reporting System* follows established software engineering standards to ensure consistency, scalability, maintainability, and ease of understanding. Adhering to these standards helps in developing a robust and well-structured system.

The system follows a **modular design approach**, where the frontend, backend, and database are separated into distinct layers. This separation of concerns improves maintainability and allows independent development and updates of each component. The backend is organized using a structured architecture with controllers, routes, middleware, and configuration files, ensuring clean code organization.

Standard **coding conventions** are followed throughout the project. Meaningful and descriptive names are used for variables, functions, and database tables, making the code more readable and understandable. Consistent indentation, formatting, and commenting practices are maintained to improve code clarity and ease of debugging.

The **user interface design** follows consistency in layout, color schemes, and navigation patterns. This ensures a smooth and intuitive user experience, especially for users with limited technical knowledge. Forms, buttons, and input fields are designed to be simple and user-friendly.

The system also follows **RESTful API design principles**, where clear and structured endpoints are defined for communication between the frontend and backend. This improves scalability and makes the system easier to integrate with other services in the future.

In terms of **database design standards**, normalization techniques are applied to reduce redundancy and maintain data integrity. Proper use of primary keys, foreign keys, and constraints ensures consistency and reliability of stored data.

Security standards are also considered, including the use of **JWT-based authentication**, password hashing, and controlled access through role-based authorization. These measures help protect sensitive data and ensure secure system operation.

Overall, following these design standards ensures that the system is well-organized, efficient, and scalable, making it easier to maintain and extend in the future.

## 4.6 Safety and Risk Mitigation Measures

The *Voice of Village: Fund Transparency & Issue Reporting System* incorporates several safety and risk mitigation measures to ensure secure, reliable, and efficient operation.

These measures are essential to protect user data, maintain system integrity, and prevent misuse.
One of the primary measures is **secure authentication and authorization**. The system uses JWT-based authentication to verify user identity, and role-based access control ensures that users can only perform actions permitted for their role. This prevents unauthorized access to sensitive functionalities.

To protect user data, **passwords are encrypted using hashing techniques**, ensuring that sensitive information is not stored in plain text. This enhances data security and reduces the risk of data breaches.

The system also includes **input validation and error handling mechanisms** to prevent invalid or malicious data from being processed. This helps in avoiding issues such as SQL injection and ensures the stability of the application.

**Data integrity** is maintained through proper database design using primary and foreign keys, ensuring that relationships between data are consistent and reliable. Regular validation checks help prevent inconsistencies.

In terms of operational safety, the system is designed to handle errors gracefully by providing appropriate error messages and preventing system crashes. Logging mechanisms can be used to track system activities and identify potential issues.
Additionally, the platform minimizes risks of misuse by maintaining transparency in actions such as issue updates and comments, which can be tracked and monitored by administrators.

Overall, these safety and risk mitigation measures ensure that the system operates securely, reliably, and efficiently, providing a trustworthy platform for users and authorities.

# 5. Implementation

## 5.1 Technology Stack

The *Voice of Village: Fund Transparency & Issue Reporting System* is developed using a simple and efficient full-stack setup.

• **Backend**

Node.js with Express.js for building APIs and handling business logic

• **Database**

MySQL for structured data storage and management

• **Frontend**

React.js for building the user interface, along with CSS for styling and React Router for navigation

• **API Communication**

Axios for connecting the frontend and backend through HTTP requests

• **Authentication**

JWT (JSON Web Tokens) for secure login and role-based access control

•**Features**

Issue reporting module for submitting and tracking complaints Comment system for communication between users and authorities Role-based system for managing different user access levels

• **Tools**

Git and GitHub for version control

**Deployment**

The system is deployed using cloud-based platforms to ensure accessibility and ease of use.

• **Backend Deployment**

The backend can be deployed on platforms like Render or similar services, which

support Node.js applications and provide API hosting with environment configuration.

• **Frontend Deployment**

The frontend application can be deployed on platforms like Vercel, which provide fast and efficient hosting for React applications.

• **Integration**

The frontend communicates with the backend through deployed API endpoints. Environment variables are used to manage API URLs and maintain secure communication between components.

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## 5.2 Module-wise Implementation

The *Voice of Village: Fund Transparency & Issue Reporting System* is developed by dividing the application into multiple modules. Each module handles a specific functionality of the system, ensuring better organization and maintainability.

• **Authentication Module**: This module handles user registration and login. It verifies user credentials, encrypts passwords, and generates JWT tokens for secure access. It also manages role-based authentication for different users.

• **User Management Module**: This module manages user details and roles such as villager, Sarpanch, administrator, and government official. It ensures that each user has appropriate permissions within the system.

• **Issue Reporting Module**: This is the core module of the system. It allows users to submit issues with details like title, description, category, and optional images. The data is stored in the database for further processing.

• **Issue Management Module**: This module allows authorized users to view all reported issues and update their status (pending, in progress, resolved). It ensures proper tracking and resolution of issues.

• **Comment Module**: This module enables users to add comments on issues. It helps in communication between citizens and authorities and provides updates on issue progress.

• **Fund Transparency Module**: This module is designed to provide visibility into fund allocation and usage. It allows users to view how funds are utilized for development activities.

• **Database Module**: This module manages all data operations, including storing, retrieving, and updating user, issue, and comment data using MySQL.

• **API Module**: This module handles communication between frontend and backend through RESTful APIs. It ensures smooth data exchange and system functionality.

Overall, the modular approach ensures that each component of the system is well-structured, making the application easier to develop, test, and maintain.

## 5.3 Code Integration Strategy

The *Voice of Village: Fund Transparency & Issue Reporting System* follows a structured code integration strategy to ensure smooth interaction between different modules and components of the system. The integration process is designed to maintain consistency, reduce errors, and enable efficient development.

The system follows a **modular integration approach**, where each module such as authentication, issue reporting, and comments is developed independently and later integrated into the main application. This approach helps in isolating errors and simplifies debugging.

The **backend integration** is organized using a layered structure that includes routes, controllers, middleware, and database configuration. Routes handle incoming API requests, controllers process the business logic, middleware manages authentication and validation, and the database layer handles data operations. All these components are connected to ensure seamless execution of functionalities.

The **frontend and backend integration** is achieved through RESTful APIs. The frontend sends HTTP requests using Axios, and the backend responds with appropriate data.

Proper API endpoints are defined for each operation such as login, issue submission, fetching issues, and adding comments.

To maintain consistency, **environment variables** are used for configuration details such as database credentials and API URLs. This ensures flexibility and security during development and deployment.

Version control is managed using **Git and GitHub**, where code changes are tracked and integrated systematically. This helps in maintaining different versions of the project and enables collaborative development.

Testing is performed after integrating each module to ensure that all components work together correctly. Any issues identified during integration are resolved before moving to the next stage.

Overall, the integration strategy ensures that all parts of the system work cohesively, resulting in a stable and efficient application.

## 5.4 Sample Code Snippets

```
— Voice of Village Database Schema—
CREATE DATABASE IF NOT EXISTS voice_of_village;
USE voice_of_village;

-- USERS TABLE
CREATE TABLE users (
id INT AUTO_INCREMENT PRIMARY KEY,
name VARCHAR(100) NOT NULL,
email VARCHAR(100) NOT NULL UNIQUE,
password VARCHAR(255) NOT NULL,
role ENUM('villager', 'sarpanch', 'admin', 'government', 'pds_dealer') NOT NULL,
mobile VARCHAR(15),
created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

-- ISSUES TABLE
CREATE TABLE issues (
id INT AUTO_INCREMENT PRIMARY KEY,
title VARCHAR(200) NOT NULL,
description TEXT NOT NULL,
```

```
category ENUM('road', 'water', 'electricity', 'sanitation', 'other') NOT NULL,
status ENUM('pending', 'in_progress', 'resolved') DEFAULT 'pending',
image_url VARCHAR(255),
submitted_by INT NOT NULL,
created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
FOREIGN KEY (submitted_by) REFERENCES users(id)
);

-- ISSUE COMMENTS TABLE
CREATE TABLE issue_comments (
id INT AUTO_INCREMENT PRIMARY KEY,
issue_id INT NOT NULL,
user_id INT NOT NULL,
comment TEXT NOT NULL,
created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
FOREIGN KEY (issue_id) REFERENCES issues(id),
FOREIGN KEY (user_id) REFERENCES users(id)
);

-- WATER REQUESTS TABLE
CREATE TABLE water_requests (
id INT AUTO_INCREMENT PRIMARY KEY,
user_id INT NOT NULL,
village_name VARCHAR(100) NOT NULL,
ward_number VARCHAR(50),
contact_number VARCHAR(15) NOT NULL,
request_type ENUM('no_supply', 'low_pressure', 'contaminated', 'pipe_leakage',
'new_connection') NOT NULL,
description TEXT NOT NULL,
days_existing INT DEFAULT 0,
households_aU̇ected INT DEFAULT 1,
priority ENUM('low', 'medium', 'urgent') DEFAULT 'medium',
image_url VARCHAR(255),
location_landmark VARCHAR(255),
status ENUM('pending', 'in_progress', 'resolved') DEFAULT 'pending',
remarks TEXT,
created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
FOREIGN KEY (user_id) REFERENCES users(id)
);

-- ELECTRICITY REQUESTS TABLE
CREATE TABLE electricity_requests (
id INT AUTO_INCREMENT PRIMARY KEY,
user_id INT NOT NULL,
village_name VARCHAR(100) NOT NULL,
ward_number VARCHAR(50),
```

```
contact_number VARCHAR(15) NOT NULL,
consumer_number VARCHAR(50),
request_type ENUM('no_supply', 'power_cuts', 'transformer_failure', 'street_light',
'damaged_pole', 'new_connection', 'meter_issue') NOT NULL,
description TEXT NOT NULL,
days_existing INT DEFAULT 0,
households_aU̇ected INT DEFAULT 1,
priority ENUM('low', 'medium', 'urgent') DEFAULT 'medium',
image_url VARCHAR(255),
location_landmark VARCHAR(255),
pole_number VARCHAR(50),
status ENUM('pending', 'in_progress', 'resolved') DEFAULT 'pending',
remarks TEXT,
created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
FOREIGN KEY (user_id) REFERENCES users(id)
);

-- HOUSE REQUESTS TABLE
CREATE TABLE house_requests (
id INT AUTO_INCREMENT PRIMARY KEY,
user_id INT NOT NULL,
village_name VARCHAR(100) NOT NULL,
ward_number VARCHAR(50),
contact_number VARCHAR(15) NOT NULL,
aadhaar_number VARCHAR(12),
plot_number VARCHAR(50),
plot_size VARCHAR(50),
survey_number VARCHAR(50),
location_landmark VARCHAR(255),
request_type ENUM('building_plan', 'noc_gram_panchayat',
'commencement_certificate', 'completion_certificate', 'occupancy_certificate',
'water_electricity_noc', 'property_tax') NOT NULL,
construction_type ENUM('new_house', 'extension', 'renovation') NOT NULL,
floors_planned INT DEFAULT 1,
estimated_cost DECIMAL(10,2),
expected_start_date DATE,
document_url VARCHAR(255),
priority ENUM('normal', 'urgent') DEFAULT 'normal',
status ENUM('pending', 'under_review', 'approved', 'rejected') DEFAULT
'pending',
remarks TEXT,
created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
FOREIGN KEY (user_id) REFERENCES users(id)
);
```

```
-- PDS RECORDS TABLE
CREATE TABLE pds_records (
id INT AUTO_INCREMENT PRIMARY KEY,
user_id INT NOT NULL,
card_type ENUM('AAY', 'PHH', 'APL') DEFAULT 'PHH',
month VARCHAR(20) NOT NULL,
year INT NOT NULL,
rice_quantity DECIMAL(5,2) DEFAULT 0,
wheat_quantity DECIMAL(5,2) DEFAULT 0,
sugar_quantity DECIMAL(5,2) DEFAULT 0,
oil_quantity DECIMAL(5,2) DEFAULT 0,
dal_quantity DECIMAL(5,2) DEFAULT 0,
collected BOOLEAN DEFAULT FALSE,
collected_at TIMESTAMP NULL,
created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
FOREIGN KEY (user_id) REFERENCES users(id)
);

-- DONATIONS TABLE
CREATE TABLE donations (
id INT AUTO_INCREMENT PRIMARY KEY,
user_id INT NOT NULL,
amount DECIMAL(10,2) NOT NULL,
razorpay_order_id VARCHAR(255),
razorpay_payment_id VARCHAR(255),
status ENUM('pending', 'success', 'failed') DEFAULT 'pending',
message TEXT,
created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
FOREIGN KEY (user_id) REFERENCES users(id)
);

-- ANNOUNCEMENTS TABLE
CREATE TABLE announcements (
id INT AUTO_INCREMENT PRIMARY KEY,
title VARCHAR(200) NOT NULL,
content TEXT NOT NULL,
priority ENUM('low', 'medium', 'high', 'urgent') DEFAULT 'medium',
posted_by INT NOT NULL,
is_active BOOLEAN DEFAULT TRUE,
created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
FOREIGN KEY (posted_by) REFERENCES users(id)
);

-- FUND DOCUMENTS TABLE
CREATE TABLE fund_documents (
id INT AUTO_INCREMENT PRIMARY KEY,
```

title VARCHAR(200) NOT NULL,
description TEXT,

## 5.5 Coding Standards Followed

The development of the *Voice of Village: Fund Transparency & Issue Reporting System* follows standard coding practices to ensure code quality, readability, and maintainability. Adhering to these standards helps in developing a clean and well-structured application.

The project follows proper **naming conventions**, where variables, functions, and database tables are given meaningful and descriptive names. This improves code clarity and makes it easier to understand the functionality of different components.

The code is written in a **modular and structured manner**, separating concerns into different layers such as routes, controllers, middleware, and configuration files. This organization helps in maintaining the code and simplifies debugging and updates.

Consistent **indentation and formatting** are maintained throughout the codebase to improve readability. Proper spacing and alignment make the code easier to review and understand.

**Code reusability** is emphasized by creating reusable functions and components wherever possible. This reduces duplication and follows the DRY (Don't Repeat Yourself) principle.

Comments are added where necessary to explain complex logic, improving the understanding of the code for future developers or reviewers.

The project also follows **security best practices**, such as hashing passwords, validating user inputs, and protecting routes using authentication mechanisms. Version control is maintained using Git, allowing tracking of changes and ensuring proper management of the codebase.

Overall, these coding standards ensure that the system is clean, efficient, and easy to maintain, supporting future enhancements and scalability.

# 6. Testing

## 6.1 Testing Strategy

The *Voice of Village: Fund Transparency & Issue Reporting System* follows a systematic testing strategy to ensure that the application functions correctly, efficiently, and securely. Testing is carried out at different stages of development to identify and fix issues early.

The project adopts a **module-based testing approach**, where each module such as authentication, issue reporting, and comments is tested individually before integration. This helps in detecting errors at an early stage and ensures that each component works as expected.

Both **manual testing** and **API testing** are performed during development. Tools like Thunder Client are used to test backend APIs by sending requests and verifying responses. This ensures that all endpoints function correctly.

After individual module testing, **integration testing** is performed to verify that different modules work together without issues. This includes testing the interaction between frontend and backend, as well as database operations.

The system is also tested for **edge cases and error handling**, such as invalid inputs, unauthorized access, and missing data. Proper error messages and responses are verified to ensure a smooth user experience.

Additionally, **user interface testing** is conducted to ensure that the application is easy to use, responsive, and visually consistent across different screens.

Overall, the testing strategy ensures that the system is reliable, secure, and performs efficiently under different scenarios, providing a smooth experience for users.

## 6.2 Unit Testing

Unit testing is performed to verify the functionality of individual components of the *Voice of Village: Fund Transparency & Issue Reporting System*. Each module is tested independently to ensure that it works correctly before integration with other parts of the system.

In this project, unit testing mainly focuses on backend functions such as user authentication, issue submission, data retrieval, and status updates. Each API endpoint is tested separately to check whether it produces the expected output for given inputs.

The authentication module is tested to ensure proper registration, login, password validation, and token generation. The issue reporting module is tested to verify that issues are correctly stored in the database with all required details. Similarly, the comment module is tested to ensure that comments are added and retrieved accurately.

Different test cases are used, including valid inputs, invalid inputs, and boundary conditions, to ensure robustness. Error handling is also verified to confirm that appropriate responses are returned for incorrect or incomplete data.

Unit testing helps in identifying bugs at an early stage and ensures that each component of the system functions correctly in isolation, improving overall system reliability.

## 6.3 Integration Testing

Integration testing is performed to ensure that different modules of the *Voice of Village: Fund Transparency & Issue Reporting System* work together correctly as a unified system. After unit testing individual components, they are integrated and tested for proper data flow and interaction.

In this project, integration testing mainly focuses on the interaction between the frontend, backend, and database. The complete workflow—from user action in the frontend to data processing in the backend and storage in the database—is verified.

The key integration testings performed are as follows:

- **Authentication Flow Integration:** Tested complete login flow including token generation, token validation, and access to protected routes.
- **Issue Reporting Flow:** Verified that issues submitted from the frontend are correctly saved in the database and can be retrieved through APIs.
- **Issue Status Update Flow:** Checked whether authorized users can update issue status and the changes are reflected correctly in the system.

- **Comment System Integration:** Ensured that comments added by users are stored and displayed correctly for each issue.
- **Role-Based Access Control**: Verified that users can only access functionalities based on their roles and unauthorized actions are restricted.
- **Error Handling and Response Flow:** Tested how the system handles invalid requests, missing data, and unauthorized access across modules.

Overall, integration testing ensures that all modules interact smoothly and the system functions as a complete and reliable application.

## 6.4 System Testing

System testing is performed to evaluate the complete *Voice of Village: Fund Transparency & Issue Reporting System* as a whole and ensure that it meets the specified requirements. In this phase, the entire application is tested in an integrated environment to verify overall functionality, performance, and reliability.

The system is tested by simulating real-world scenarios where different users interact with the application. This includes activities such as user registration, login, issue submission, issue tracking, status updates by authorities, and commenting. The goal is to ensure that all features work together correctly and provide the expected output.

Both **functional testing** and **non-functional testing** are carried out during this phase. Functional testing ensures that all system features operate as intended, while non-functional testing focuses on aspects such as performance, usability, and security.

The key system testings performed include:

- **End-to-End Testing**: Verified the complete workflow from user login to issue resolution, ensuring smooth execution of all processes.
- **User Role Testing**: Ensured that different users (villager, Sarpanch, admin, government official) can access only their respective functionalities.
- **Performance Testing**: Checked system response time and behavior under multiple requests to ensure efficient performance.

- **Usability Testing**: Verified that the interface is user-friendly, easy to navigate, and suitable for users with minimal technical knowledge.
- **Security Testing**: Tested authentication, authorization, and data protection mechanisms to ensure secure system operation.
- **Error Handling Testing**: Ensured that the system handles errors gracefully and provides appropriate messages to users.

Overall, system testing confirms that the application is stable, reliable, and ready for deployment, providing a seamless experience to all users.

## 6.5 Test Cases and Results

Multiple test cases were designed and executed to validate the core functionalities of the EcoTrace system under different scenarios. The testing covered authentication, product and batch management, AI evaluation, transport tracking, lab verification, and QR-based access.

| Test Case | Input Scenario | Expected Result | Actual Result | Status |
|---|---|---|---|---|
| User Registration | Valid email and password | Account created successfully | Account created | Passed |
| User Registration | Duplicate email | Error message shown | Error displayed | Passed |
| User Login | Correct credentials | Login successful, token generated | Login successful | Passed |
| User Login | Incorrect password | Access denied | Error displayed | Passed |
| Submit Issue | Valid product data | Product created and stored | Product created | Passed |
| Get All Issues | Valid materials and percentages | Batch created successfully | Batch created | Passed |
| Update Issue Status | No change in materials | Auto verification | Verified automatically | Passed |
| Add Comment | Minor change in composition | AI scoring triggered | Score generated | Passed |
| Unauthorized Access | Major change | Lab verification required | Marked for lab | Passed |

| | | | | |
|---|---|---|---|---|
| Role-Based Access | Valid batch data | Score and insights generated | Score displayed | Passed |
| Error Handling | Valid report data | Report submitted | Report created | Passed |
| Admin Approval | Approve lab report | Batch verified | Verified | Passed |
| Admin Rejection | Reject lab report | Batch rejected | Rejected | Passed |

**Table 6.5.1 Test Cases and Results**

## 6.6 Bug Reporting and Tracking

Bug reporting and tracking is an important part of the development process to ensure the reliability and quality of the *Voice of Village: Fund Transparency & Issue Reporting System*. During testing, any issues or defects identified in the system are recorded, analyzed, and resolved systematically.

Bugs are identified during unit testing, integration testing, and system testing phases. Each bug is documented with details such as description, module affected, steps to reproduce, and expected versus actual results. This helps in understanding the issue clearly and fixing it efficiently.

Common types of bugs encountered include validation errors, incorrect API responses, authentication issues, and data handling problems. These issues are tracked and resolved during development to ensure smooth system functionality.

Version control tools like Git are used to manage changes and track bug fixes. Each fix is updated in the codebase with proper commits, making it easier to maintain a history of changes.

After fixing a bug, the system is retested to ensure that the issue is resolved and does not affect other functionalities. This process helps in maintaining system stability and improving overall performance.

Overall, effective bug reporting and tracking ensures that the system is robust, reliable, and free from critical errors before deployment.

## 6.7 Quality Assurance Standards

Quality assurance is an essential part of the *Voice of Village: Fund Transparency & Issue Reporting System* to ensure that the application meets required standards of performance, reliability, and usability

The project follows a **systematic testing approach**, including unit testing, integration testing, and system testing, to verify that all functionalities work correctly. Each module is tested individually and then as part of the complete system to ensure consistency.

**Code quality standards** are maintained by following proper coding conventions, meaningful naming, and modular design. This improves readability, maintainability, and reduces the chances of errors.

**Security standards** are followed by implementing authentication and authorization mechanisms using JWT. Passwords are securely hashed, and access to protected routes is restricted based on user roles. **Usability standards** are maintained by designing a simple and intuitive user interface. The application is tested to ensure smooth navigation and ease of use for all types of users. Regular **bug tracking and resolution** are carried out to identify and fix issues during development. Retesting is performed after fixing bugs to ensure system stability.

Overall, these quality assurance practices ensure that the system is reliable, secure, user-friendly, and ready for deployment.

# 7. Results and Discussion

## 7.1 Output Screenshots

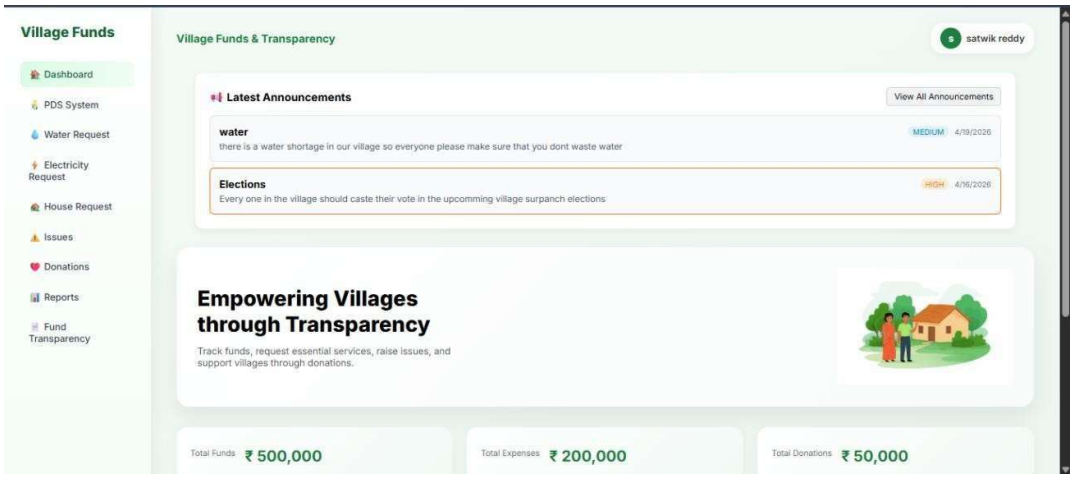


**Fig 7.1.1 Villager**

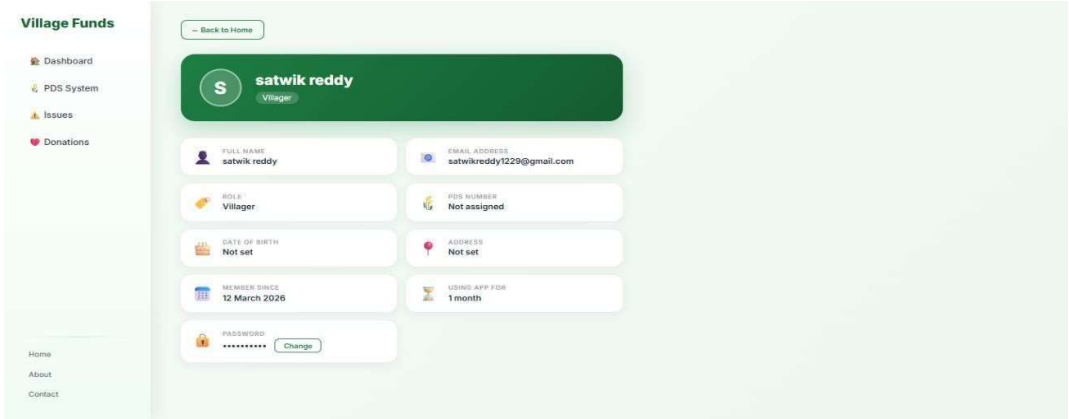


**Fig 7.1.2 villager profile**

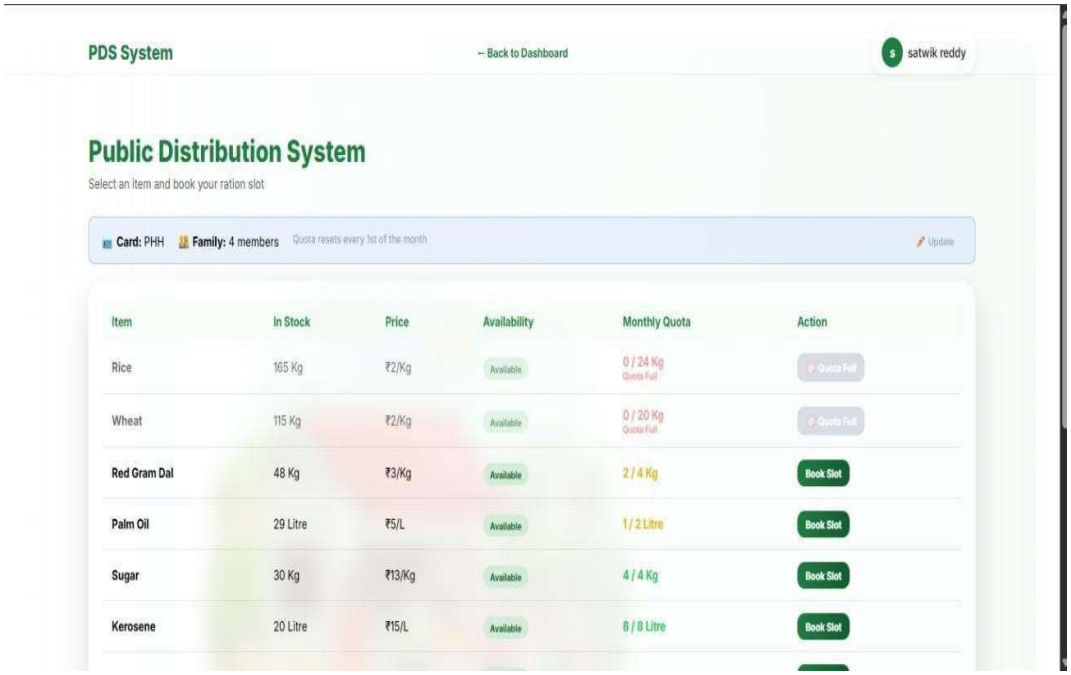


**FIG 7.1.3.1 PDS System**

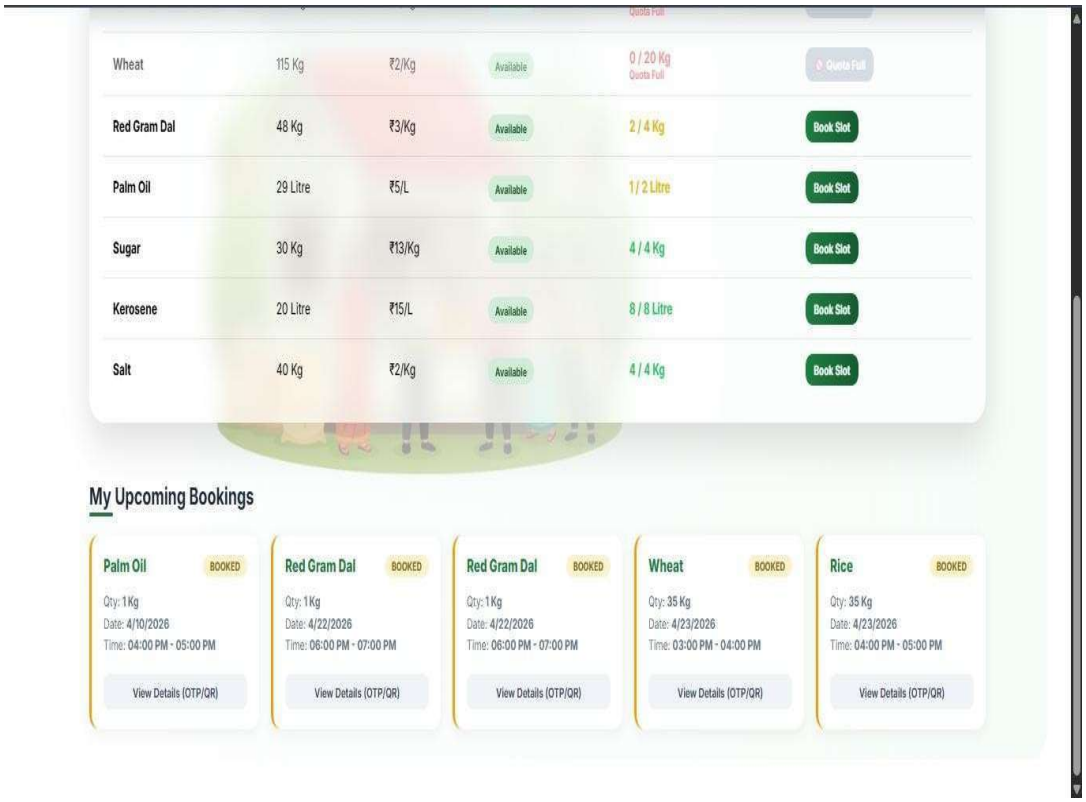


**Fig 7.1.3.2 PDS System**

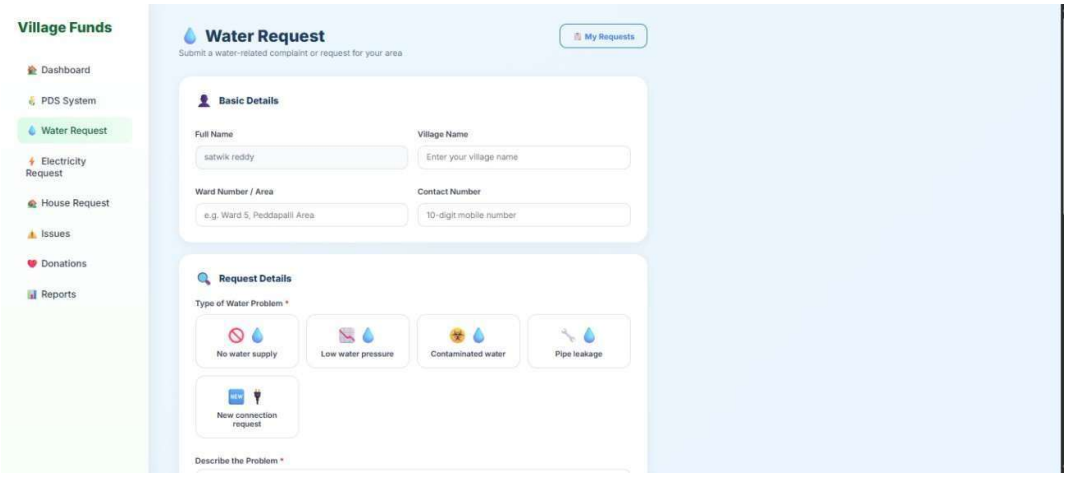


**Fig 7.1.4.1 water request**

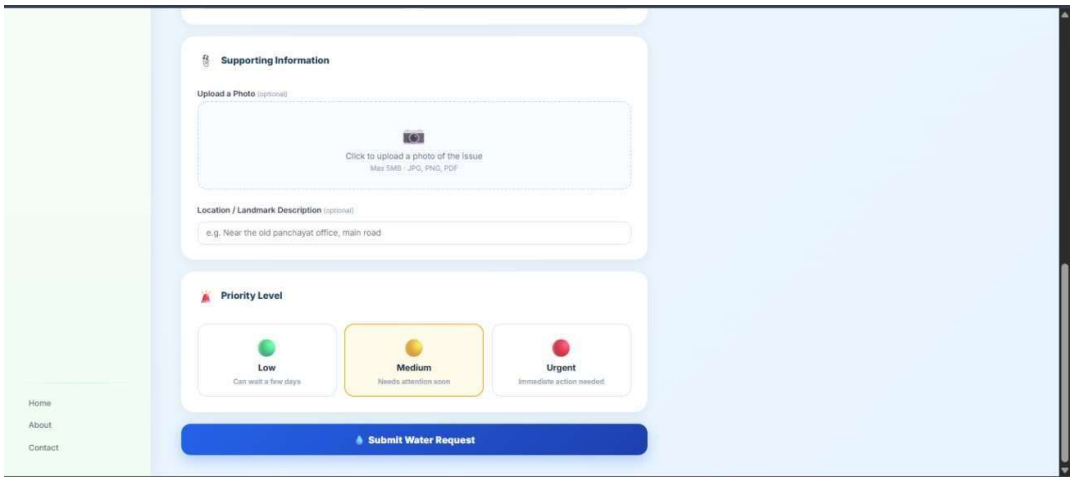


**Fig 7.1.4.2 water request**

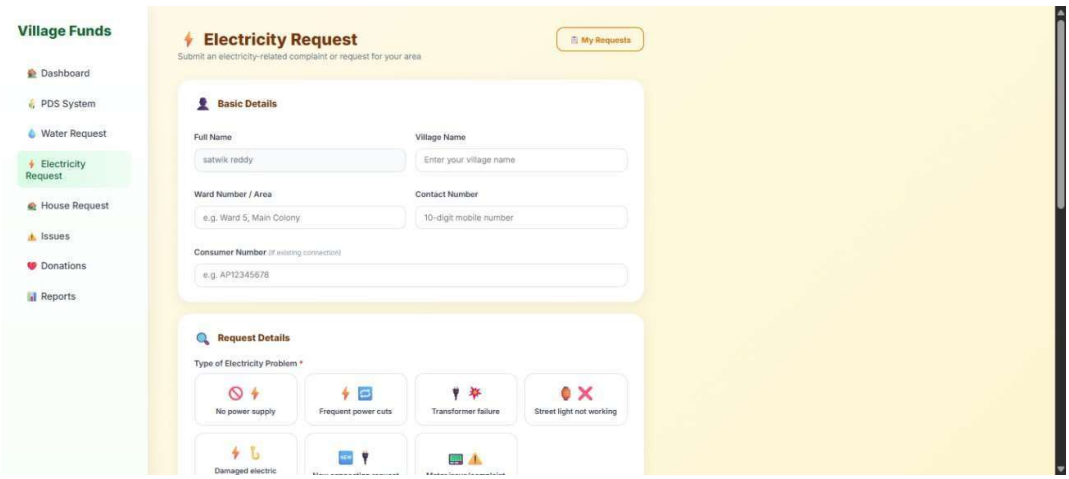


**Fig 7.1.5.1 Electricity request**

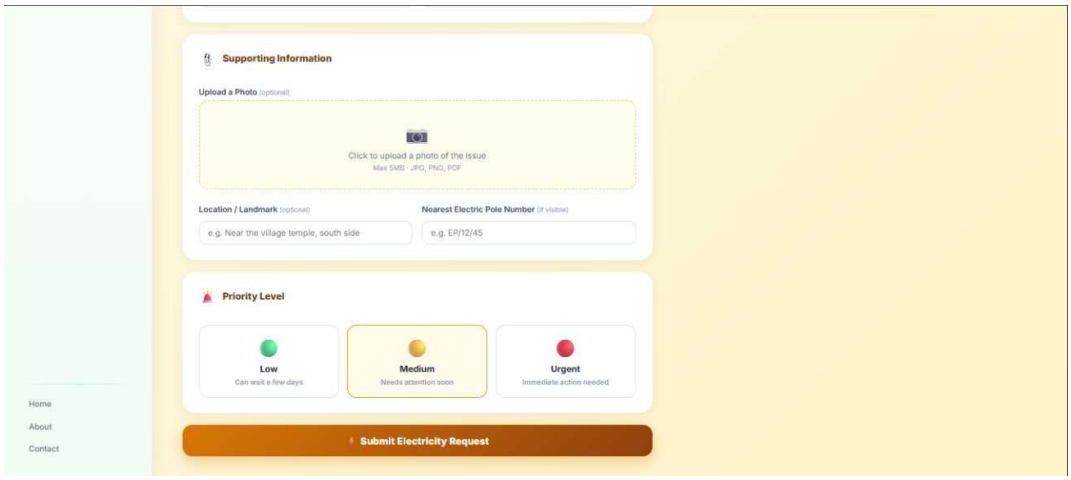


**Fig 7.1.5.2 Electricity request**

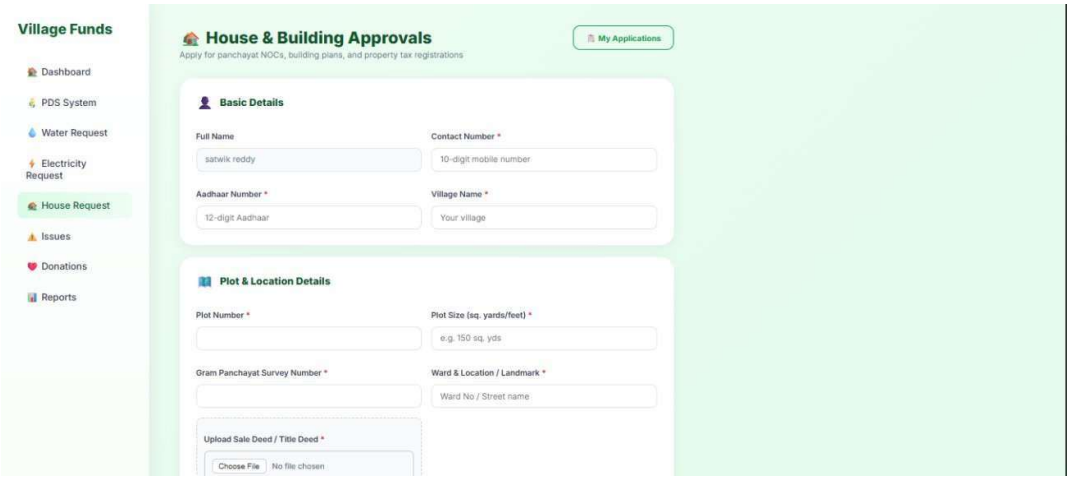


**Fig 7.1.6.1 House request**

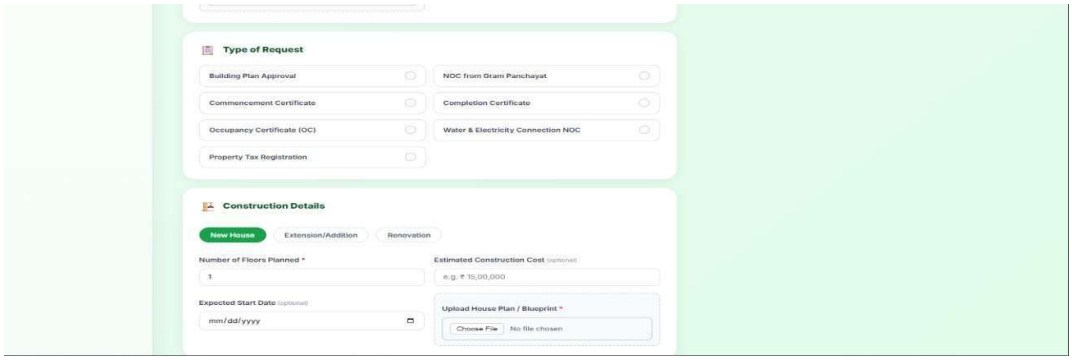


**Fig 7.1.6.2 House Request**

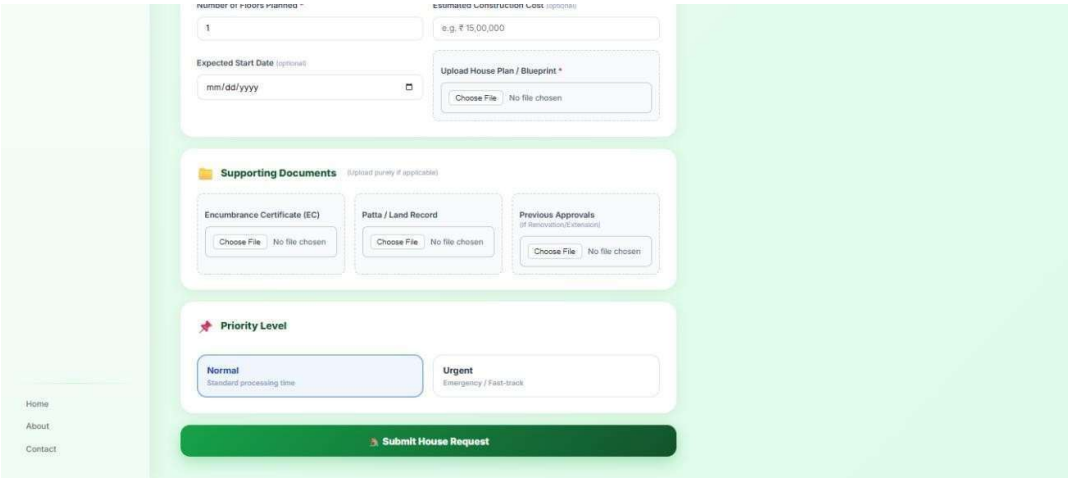


**Fig 7.1.6.3 House request**

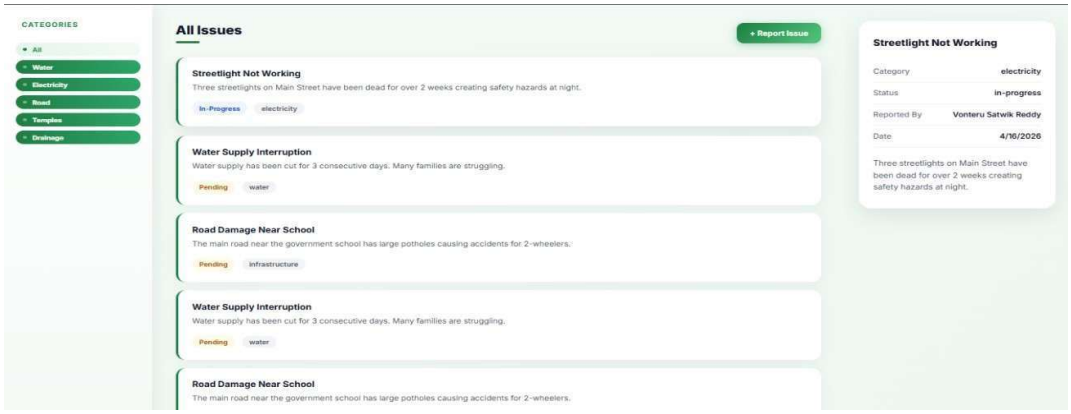


**Fig 7.1.7 Issues Reporting**

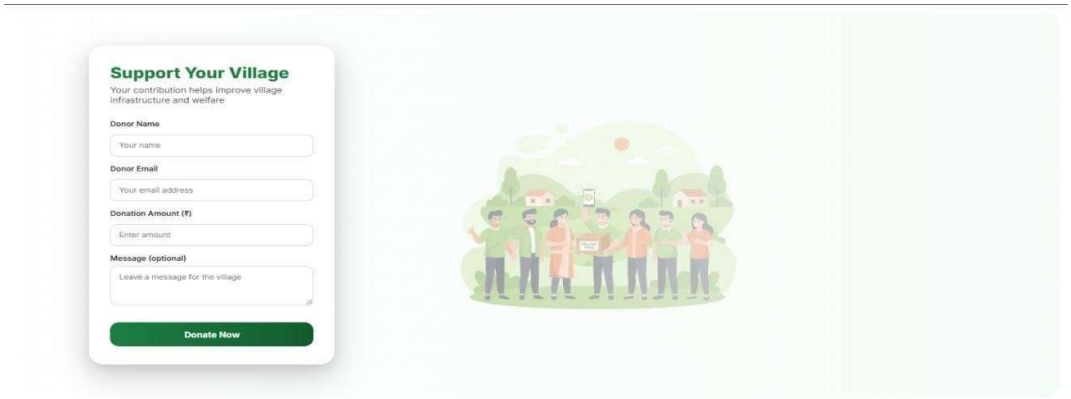


**Fig 7.1.8 Donations page**

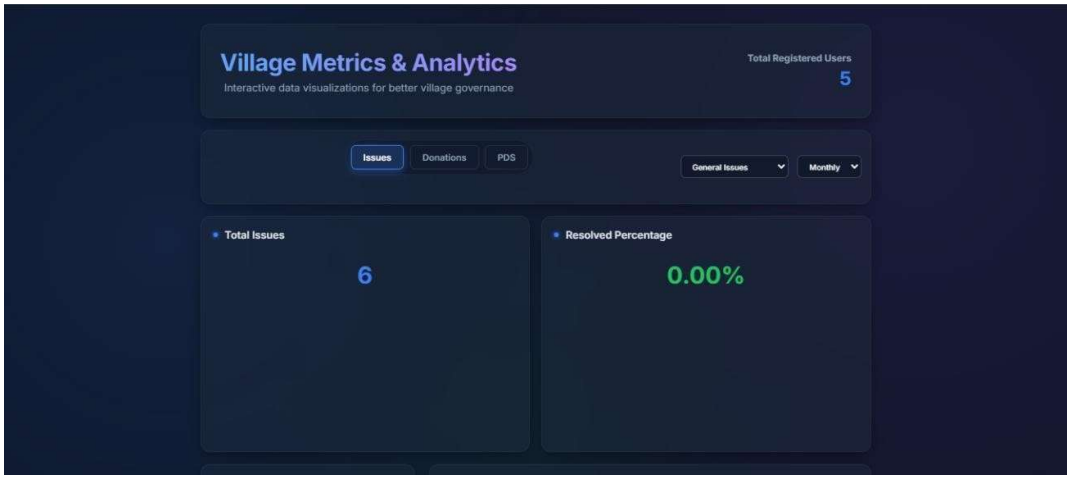


**Fig 7.1.9 Reports**

**Fig 7.1.8.2 Reports**

## 7.2 Results Interpretation

The results of the *Voice of Village: Fund Transparency & Issue Reporting System* demonstrate that the application successfully achieves its objective of improving transparency and communication in village governance.

The system enables users to report issues efficiently and track their progress in real time. This ensures that citizens are aware of the status of their complaints, which increases trust and engagement. The issue management functionality allows authorities to update and resolve issues in a structured manner, reducing delays and miscommunication.

The implementation of role-based access control ensures that different users can perform only their designated tasks, improving system security and organization. The comment feature enhances interaction between citizens and authorities, making the process more transparent and collaborative.

The integration between frontend, backend, and database works smoothly, ensuring accurate data flow and reliable system performance. The testing results confirm that all major functionalities operate correctly under different scenarios.

Overall, the system provides an effective solution for digitizing village-level governance, improving efficiency, accountability, and user participation.

## 7.3 Performance Evaluation

The performance of the *Voice of Village: Fund Transparency & Issue Reporting System* is evaluated based on factors such as response time, system reliability, and efficiency of operations. The system demonstrates satisfactory performance under normal usage conditions.

The application provides **fast response times** for most operations such as user login, issue submission, and data retrieval. The use of efficient backend processing and structured database queries ensures that requests are handled quickly.

The system shows good **reliability**, as it performs consistently without major failures during testing. Error handling mechanisms ensure that unexpected inputs or issues do not cause system crashes.

In terms of **efficiency**, the system is capable of handling multiple user requests simultaneously without significant delays. The modular design and optimized API communication contribute to smooth performance.

The **frontend interface** is responsive and provides a seamless user experience across different devices. Pages load quickly, and navigation between features is smooth.

Overall, the system meets the expected performance requirements and provides a stable and efficient platform for users. Further optimization can be done in the future to support larger-scale deployment and increased user load.

## 7.4 Comparative Results

The performance and effectiveness of the *Voice of Village: Fund Transparency & Issue Reporting System* were evaluated by comparing it with existing approaches used in village-level governance.

In traditional systems, issue reporting is mostly manual, leading to delays, lack of tracking, and poor communication. In contrast, the proposed system provides a **digital and structured platform** where issues can be reported and tracked in real time, significantly improving efficiency.

Existing systems often lack transparency in processes and fund utilization. The proposed system enhances **transparency and accountability** by allowing users to monitor issue

status and view relevant updates, thereby increasing trust between citizens and authorities.
Communication in traditional methods is limited and unstructured, whereas the system introduces a **comment-based interaction mechanism**, enabling better coordination and clarity among stakeholders.

Additionally, the use of **role-based access control** ensures proper management of responsibilities, which is generally absent in existing approaches. This leads to better organization and reduced chances of misuse.
Overall, the proposed system outperforms existing methods by providing improved efficiency, transparency, communication, and user engagement, making it a more effective solution for village governance.

# 8. Conclusion and Future Scope

## 8.1 Summary of Work Done

The *Voice of Village: Fund Transparency & Issue Reporting System* was developed to address the challenges of transparency, accountability, and communication in village-level governance. The project focuses on providing a digital platform that enables citizens to report issues and track their resolution while also promoting transparency in administrative processes.

The system was designed and implemented using a full-stack approach. The frontend was developed using React.js to provide an interactive and user-friendly interface, while the backend was built using Node.js and Express.js to handle business logic and API communication. MySQL was used as the database for storing and managing structured data such as users, issues, and comments.

The project was developed in multiple phases, starting with requirement analysis and system design, followed by implementation of core modules such as user authentication, issue reporting, issue management, and commenting. Role-based access control was implemented to support different types of users including villagers, Sarpanch, administrators, and government officials.

Various testing techniques, including unit testing, integration testing, and system testing, were performed to ensure that the application functions correctly and efficiently. Bugs identified during testing were resolved, and quality assurance practices were followed to maintain system reliability and performance.

The final system successfully demonstrates a working solution for digital governance at the village level. It provides a centralized platform for issue reporting, real-time tracking, and improved communication between citizens and authorities.

Overall, the project achieves its objective of creating a transparent, efficient, and user-friendly system that enhances participation and accountability in village development.

## 8.2 Limitations

While the *Voice of Village: Fund Transparency & Issue Reporting System* provides an effective solution for improving transparency and issue management, it has certain limitations.

The system is developed for a **controlled environment** and is not integrated with real-time government databases or external administrative systems. This limits its ability to reflect actual government-level data and operations.

The **fund transparency module is conceptual and basic**, and does not include advanced features such as detailed financial analytics, auditing mechanisms, or real-time fund tracking.

The application currently depends on **manual updates by authorities** for issue status, which may lead to delays if updates are not performed regularly.

The system requires **internet connectivity**, which may be a challenge in rural areas with limited or unstable network access.

Advanced features such as **real-time notifications, mobile application support, and geolocation-based issue tracking** are not included in the current implementation.

The system is designed for **small to medium-scale usage** and may require further optimization and infrastructure support for large-scale deployment.

Despite these limitations, the system provides a strong foundation and can be enhanced in the future to overcome these challenges.

## 8.3 Challenges Faced

During the development of the *Voice of Village: Fund Transparency & Issue Reporting System*, several challenges were encountered at different stages of the project. These challenges helped in gaining practical experience and improving problem-solving skills.

One of the major challenges was **understanding and designing the system architecture**, especially in integrating frontend, backend, and database components. Ensuring smooth communication between these layers required careful planning and implementation.

Another challenge was **implementing authentication and role-based access control**. Managing different user roles such as villager, Sarpanch, administrator, and government official while ensuring secure access to specific functionalities required proper logic and testing.

**Database design and management** also posed challenges, particularly in establishing relationships between tables and maintaining data consistency. Writing efficient queries and handling data operations correctly required attention to detail.

During development, **API integration between frontend and backend** was a key challenge. Handling API requests, responses, and error cases while ensuring proper data flow required debugging and testing.

The team also faced issues related to **debugging and error handling**, especially when dealing with unexpected inputs or server-side errors. Identifying and resolving bugs efficiently was an important part of the development process.

Another challenge was **time management and coordination**, as the project involved multiple modules that needed to be completed within a limited timeframe.

Additionally, ensuring a **user-friendly interface** that is simple and accessible for all users required multiple iterations and improvements.

Despite these challenges, they were successfully addressed through continuous learning, testing, and teamwork, leading to the successful completion of the project.

## 8.4 Future Enhancements

- The *Voice of Village: Fund Transparency & Issue Reporting System* provides a strong foundation for improving transparency and communication in village governance. However, there are several enhancements that can be implemented in the future to extend its functionality and effectiveness.
- One of the major enhancements is the development of a **mobile application**, which would make the system more accessible to users, especially in rural areas where mobile usage is more common than desktops.
- The system can be improved by adding **real-time notifications**, such as SMS or push notifications, to inform users about issue updates, status changes, and important announcements.
- Integration with **government databases and e-governance systems** can be implemented to provide real-time and accurate data related to fund allocation and administrative actions.
- The **fund transparency module** can be enhanced with advanced features such as detailed financial reports, visual analytics, and auditing mechanisms to provide deeper insights into fund utilization.
- Incorporating **geolocation features** can help users report issues with location data, making it easier for authorities to identify and resolve problems efficiently.
- The system can also include **AI-based analytics** to prioritize issues based on severity, frequency, or impact, helping authorities make better decisions.
- Improving **security features**, such as multi-factor authentication and advanced data protection mechanisms, can further enhance system safety.
- Additionally, the system can be scaled for **larger deployments**, supporting multiple villages or regions with centralized monitoring and management.
- Overall, these enhancements can transform the system into a more powerful, scalable, and real-world-ready solution for digital governance

# 9. References

These sources include official documentation, tutorials, and technical resources that helped in understanding and implementing various technologies.


**Technical Publications**

- Pressman, R. S., *Software Engineering: A Practitioner's Approach*
- Sommerville, I., *Software Engineering*

**Websites and Documentation**

- Node.js Official Documentation – https://nodejs.org
- Express.js Documentation – https://expressjs.com
- React Documentation – https://react.dev
- MySQL Documentation – https://www.mysql.com
- JWT Introduction – https://jwt.io
- Railway databse- https://railway.com/

**Tools and Platforms**

- Git – https://git-scm.com
- GitHub – https://github.com
- Postman / Thunder Client for API testing

**Other Sources**

- Online tutorials and developer forums such as Stack Overflow
- YouTube tutorials related to full-stack web development


These references provided guidance for designing, developing, and testing the system effective

# 10. Appendices

## A. SDLC Forms

This section includes the Software Development Life Cycle (SDLC) related documents used during the development of the *Voice of Village: Fund Transparency & Issue Reporting System*. These forms help in documenting requirements, design decisions, and testing activities in a structured manner.

**1. Software Requirements Specification (SRS):** The SRS document defines the overall requirements of the system, including functional and non-functional requirements, user roles, and system constraints. It serves as a blueprint for system design and development.

**2. Feasibility Study Report:** This document analyzes the technical, economic, and operational feasibility of the project. It ensures that the system is practical, cost-effective, and suitable for implementation.

**3. Design Documents:** These include system architecture diagrams, database design (ER diagram), and UML diagrams such as use case, class, sequence, and activity diagrams. These documents provide a clear understanding of the system structure and workflow.

**4. Test Reports:** Test reports contain details of test cases, test results, and bug tracking. They help in verifying that the system functions correctly and meets the specified requirements.

**5. Requirement Analysis Documents:** These documents capture the initial requirements gathered from the problem statement and user needs. They guide the overall development process.

All these SDLC forms ensure that the project follows a structured development process and maintains proper documentation throughout its lifecycle.

**B. Gantt Chart**

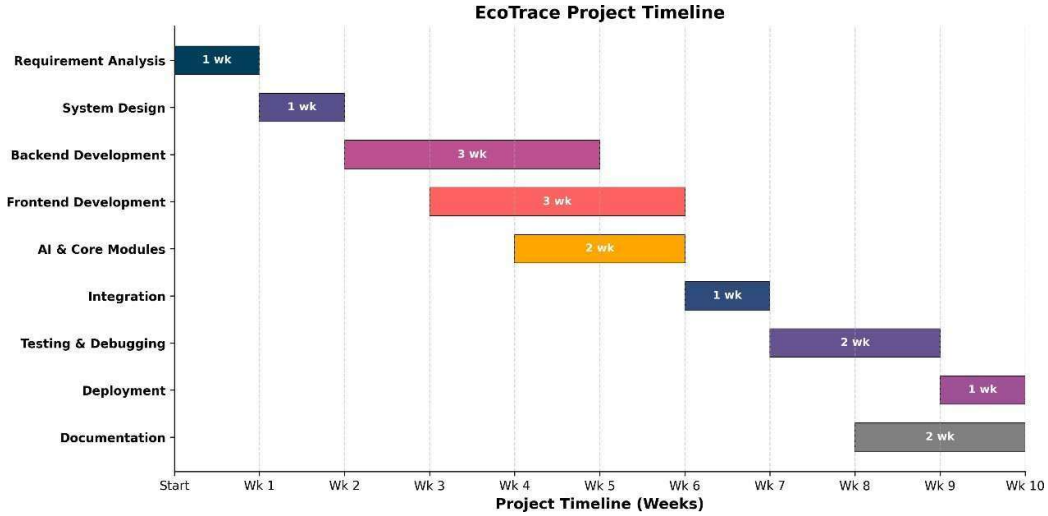


**Fig 10.2 Gantt Chart**

## C. Ethical Considerations and Consent

The development of the *Voice of Village: Fund Transparency & Issue Reporting System* considers important ethical aspects to ensure responsible and fair usage of the platform. The system is designed to promote transparency, accountability, and user trust while protecting user data and privacy.

One of the key considerations is data privacy and confidentiality. Personal information such as user details and reported issues is handled securely and is not shared with unauthorized parties. Appropriate measures such as authentication and access control are implemented to protect sensitive data.

The system ensures fair and unbiased usage, where all users have equal access to report issues and view information. No discrimination is made based on user role beyond functional permissions required for system operation.

User consent is considered during registration, where users agree to provide their information for system usage. Users are informed about how their data will be used within the platform.

The platform also promotes transparency and accountability, ensuring that actions such as issue updates and comments are visible and traceable. This reduces the chances of misuse and encourages responsible behavior among users and authorities.

Additionally, the system is designed to avoid misuse by implementing restrictions on unauthorized access and improper activities.

Overall, these ethical considerations ensure that the system operates in a responsible, secure, and user-centric manner while maintaining trust among all stakeholders.

**D. Plagiarism Report**

**E. Source Code Repository and Deployment Links**

The source code repository is maintained using GitHub, where all project files, including frontend, backend, and database-related configurations, are stored. Version control is used to track changes, manage updates, and maintain the project history.

The deployment links provide access to the live application (if deployed), allowing users to interact with the system in a real-time environment. These links include both frontend and backend endpoints.

GitHub Repository Link: https://github.com/Satwik1229/Voice-Of-Village

**F. Soft Copy / Drive Link for all project related files**

Google Drive Link:
https://drive.google.com/drive/folders/1PhPBj0fZxbn6AwU2r0nR3BNxFDfQjhKW

Deployment Link: https://voice-village-host--udayyadav200323.replit.app

Overall, the project provides a technological approach to achieving sustainable development by improving governance, ensuring transparency in fund utilization, and strengthening communication between citizens and authorities.

**G. Journal / Conference paper published on project**

IRJMETS

International Research Journal Of Modernization in Engineering Technology and Science

(Peer-Reviewed, Open Access, Fully Refereed International Journal)

e-ISSN: 2582-5208

Ref: IRJMETS/Certificate/Volume 08/Issue 04/80400095379

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